The Fertility, Marriage, and Gender Equality Quandary

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Abstract

This paper documents a three-way trade-off: simultaneously achieving high fertility, low single parenthood, and gender income equality is rare across countries and over time. A mutual information analysis shows that the three outcomes exhibit *synergy*, meaning that the dependence between any two strengthens once the third is conditioned on. An equilibrium marriage market model rationalizes the pattern and shows that, while standard policies each improve two outcomes at the expense of the third, *redistributing* childcare from wives to husbands can raise all three under certain conditions. Calibrating the model to Mexico over 1990 to 2015, I quantify the contributions of gender-neutral and gender-biased technological changes.

JEL classification: D13, J11, J12, J13, J16

Keywords: Gender equality, family structure, fertility

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1 Introduction

Three demographic transformations have reshaped advanced economies over the past half-century. Fertility has fallen, often well below replacement. Stable dual-parent families have become less common, and rising shares of children now grow up in single-parent households. Gender gaps in education and earnings have narrowed substantially. Together, these shifts constitute the grand gender convergence described by Goldin (2014).

Each trend has been carefully studied on its own, and pairs of trends have been studied as well, including fertility with female labor supply, marriage with female earnings, and marriage with fertility. Much less is known about how the three outcomes move together. This paper asks whether, once all three are examined jointly, a pattern emerges that is invisible in any pairwise comparison.

The starting point is a simple empirical observation. Across four datasets—an OECD panel covering 1970 to 2014, a global panel of 95 countries from 1990 to 2015, the fifty U.S. states in 2023, and 741 U.S. commuting zones in 2000—societies that succeed on any two of high fertility, low single parenthood, and gender income equality almost invariably fall short on the third. The share of observations achieving all three ranges from 2.5 to 7.6 percent, well below the 12.5 percent benchmark implied by statistical independence. This scarcity is not a mechanical consequence of the familiar pairwise trade-offs. Using mutual information analysis from information theory, I show that the three outcomes exhibit *synergy*: the dependence between any two of them *strengthens* once the third is conditioned on. The three variables jointly carry information that no pair of them carries alone.

Building on this empirical motivation, the paper makes three contributions. The first is to document the three-way scarcity as a robust stylized fact. The pattern appears in cross-country, cross-state, and cross-commuting-zone data; it holds in both recent cross sections and multi-decade panels; it survives alternative measures of family structure, whether out-of-wedlock births or children's co-residence with both biological parents; and it persists when samples are stratified by income and education.

The second contribution is a tractable equilibrium marriage market model that rationalizes the trade-off through the interaction of three mechanisms rather than through any single pairwise channel. Men and women draw idiosyncratic taste shocks for marriage and match competitively, but the price that clears the marriage market is not a scalar. It is a two-dimensional vector consisting of an income transfer α from husbands to wives and marital fertility n^m. Because wives bear a time cost for children, fertility and female labor supply are linked endogenously, and because marriage bundles income-sharing with shared fertility, the gender wage gap maps directly into both the gender income gap and the marriage rate. The model thus knits together three building blocks familiar from prior work, including fertility-and-labor-supply, transferable-utility matching, and bargaining over fertility, and it shows that their joint operation produces a three-way tension that no two of them can generate in isolation.

The third contribution is to identify a policy lever capable of relaxing the trade-off. Three standard instruments, namely subsidizing marriage, closing the wage gap, and lowering the overall cost of children, each improve two of the three outcomes while worsening the third, so the trade-off is genuinely binding under conventional policy. A policy that instead *redistributes* childcare from wives to husbands can, under two transparent conditions, raise fertility, marriage, and gender income equality simultaneously. The first condition requires that the wife's earnings exceed the income transfer she receives from her husband. The second condition requires that the husband's utility gain from higher fertility outweighs his consumption loss from additional childcare time, and it pinpoints the parameter region in which the policy delivers a Pareto improvement within the household.

The logic behind this result is worth stating plainly. In the baseline model, the asymmetric allocation of childcare is taken as given, reflecting the well-documented empirical pattern that mothers shoulder far more of the time burden than fathers. Viewed through a wider lens, however, this asymmetry is itself plausibly a legacy of traditional gender norms rather than a primitive of technology or preferences. When the norm concentrates childcare on women, the household's allocation of time is distorted away from what the underlying fundamentals would dictate. Redistribution can therefore be understood as a corrective that realigns time allocation with comparative advantage, and the resulting gains can be large enough to loosen what otherwise appears to be a binding constraint.

The two conditions also delineate *where* the policy delivers these gains. They are most likely to hold jointly in settings with moderate female wages, a strong preference for children, and a low baseline paternal childcare share—features that characterize many middle-income economies, including the Mexican calibration studied below. They are less likely to hold in high-income, low-fertility economies, where fathers already contribute more at baseline and the fertility response to redistribution is muted. This pattern is consistent with the mixed evidence on reserved paternity-leave quotas, which has narrowed within-household gender gaps across high-income settings but produced heterogeneous fertility responses.

The takeaway is sharp but conditional. Instruments that move *levels* of wages, marriage incentives, or the overall cost of children trade one outcome off against the others. Instruments that move the *intra-household allocation of time*, by dislodging a norm-driven asymmetry, can loosen the trade-off altogether—but only in the region of the parameter space where both conditions bind, a region that the framework makes precise and that plausibly includes middle-income settings underrepresented in the existing empirical literature.

I then bring the framework to the data by calibrating it to Mexico from 1990 to 2015. The period spans a halving of fertility, a ten-percentage-point rise in single motherhood, a complete reversal of the gender education gap, and a substantial narrowing of gender earnings gaps. With estimated parameters, the model tracks Mexico's trajectory for both fertility and dual parenthood, and its 2023 out-of-sample fertility prediction of 1.93 is within 0.02 of the realized value of 1.91. A structural decomposition attributes roughly half of the fertility and dual-parenthood declines to gender-neutral productivity growth and most of the gender income convergence to declining gender-biased productivity and human-capital convergence. Both conditions for redistribution to be Pareto-improving hold throughout the sample, which suggests that paternity-leave mandates with reserved father quotas, of the kind implemented in Sweden and Iceland, could be welfare-improving in environments of this type.

A dynamic extension closes the paper by introducing a feedback through which lower marriage rates depress future male human capital disproportionately, consistent with evidence that boys are more sensitive than girls to the absence of a biological father. The feedback amplifies the direct effect of any technological shock, so a single change in productivity cascades into self-reinforcing declines in marriage and fertility across generations.

Related Literature

The paper bridges several literatures that have developed largely in parallel. A long tradition links fertility decisions to female labor supply through the time cost of children, beginning with Rosenzweig and Wolpin (1980) and developed in modern quantitative form by Erosa et al. (2016) and Blundell et al. (2016). Related work emphasizes household production and the marketization of childcare (Bar et al., 2018; Greenwood, 2019) as well as parental leave (Kim and Yum, 2025), with comprehensive recent surveys provided by Doepke et al. (2023) and Greenwood et al. (2023). My contribution is not to refine the fertility-and-labor-supply margin but to embed a stylized version of it in a marriage market in which fertility, marriage, and the gender income gap are jointly determined. Richer life-cycle treatments would only sharpen the trade-off through human-capital depreciation.

A second literature, originating with Becker (1973) and developed in transferable-utility frameworks by Choo and Siow (2006), Chiappori (2017), and Low (2024), studies how marriage markets clear when men and women bring different productivities and preferences. Quantitative analyses of marriage and divorce appear in Stevenson and Wolfers (2007), Voena (2015), Fernández and Wong (2014), Santos and Weiss (2016), Greenwood et al. (2016), and Kennes and Knowles (2024). I abstract from within-gender heterogeneity to focus on the between-gender margin, and I extend the standard transferable-utility setup by allowing a two-dimensional clearing price consisting of an income transfer plus shared fertility. This extension is essential for generating the three-way tension that a scalar transfer cannot produce.

A third strand emphasizes that fertility decisions inside marriage reflect bargaining between spouses with different preferences (Doepke and Tertilt, 2018; Doepke and Kindermann, 2019; Myong et al., 2021), and a related body of work documents how women's bargaining power has evolved with social and legal change (Doepke and Tertilt, 2009; Folbre, 2021). I adopt the wife's-veto assumption from Doepke and Kindermann (2019) as the simplest device for letting wives' preferences pin down marital fertility, and I embed it in a marriage market with two-dimensional pricing so that the wife's choice of n^m enters both intra-household allocations and the equilibrium incentives to marry.

A fourth literature documents that the absence of a biological parent harms children's human capital and that boys appear more vulnerable than girls (Sommers, 2001; Bertrand and Pan, 2013; Chetty et al., 2016; Autor et al., 2019; Wasserman, 2020; Reeves, 2022; Autor et al., 2023; Frimmel et al., 2024), with Kearney (2023) drawing out the implications for inequality and mobility. My dynamic extension uses this evidence to motivate a feedback from current marriage rates to future male human capital, which generates self-reinforcing declines in marriage and fertility under technological progress.

A fifth, more macroeconomic literature studies how technological change has reshaped the family (Greenwood et al., 2005a,b, 2016; Ngai and Petrongolo, 2017; Kuhn et al., 2024; Cao et al., 2024), with Olivetti and Petrongolo (2016) and Feng et al. (2023) documenting the long-run evolution of gender gaps and Hsieh et al. (2019) accounting for the aggregate gains from improved talent allocation. I borrow from this tradition the device of decomposing wage growth into gender-neutral and gender-biased components, but I depart from it by tying the three outcomes together explicitly through marriage-market equilibrium. The same technological shock affects fertility, family structure, and gender gaps through interconnected channels rather than parallel ones. On the empirical side, the paper combines the dartboard logic of Ellison and Glaeser (1997), which gives meaning to scarcity in a joint distribution relative to a randomness benchmark, with mutual information analysis from information theory to test whether the three outcomes are merely *redundant*, in which case pairwise analyses would suffice, or genuinely *synergistic*, in which case they cannot. The finding of synergy provides the formal motivation for the joint equilibrium framework that follows.

The remainder of the paper proceeds as follows. Section 2 documents the three-way tradeoff. Section 3 develops the model, characterizes equilibrium, and proves the comparative-statics results, including the redistribution proposition. Section 4 calibrates the model to Mexico and presents the decomposition. Section 5 introduces the dynamic feedback through male human capital. Section 6 concludes.

2 Motivating Facts

This section establishes the three-way tension that motivates the theory. I use two complementary statistical tools. Venn diagrams show how often all three outcomes are achieved together, and mutual information analysis tests whether the three outcomes must be studied jointly rather than as three separate pairs.

2.1 Data and Methods

Data sources and samples

Four datasets, summarized in Table 1, provide complementary views of the same phenomenon. Each dataset serves a specific purpose, and the triangulation across them is part of the argument.

Sample	N	Period	Family Structure	Gender Equality
OECD	721	1970–2014	Out-of-wedlock births	Median earnings gap
World	1,130	1990–2015	Children with both parents	Female income share
U.S. States	51	2023	Single-mother households	Median earnings gap
U.S. CZ	741	2000	Single-mother households	Earnings ratio

Table 1. Summary of Data Samples

Notes: CZ = Commuting Zones. All samples use the total fertility rate for fertility. OECD data are drawn from the OECD Family Database. The World sample combines UN fertility data, children's living arrangements from Brenøe and Wasserman (2025), and the female labor income share from the World Inequality Database (Chancel et al., 2022). U.S. state data come from CDC Natality files and the American Community Survey. U.S. CZ data are from Chetty et al. (2014).

The *OECD sample* spans 37 countries from 1970 to 2014 and provides the longest time dimension. It traces country trajectories over the full postwar period, at the cost of using out-of-wedlock births as a proxy for absent fathers. This proxy is imperfect. In Nordic Europe in particular, children increasingly live with both biological parents in stable non-marital unions, so out-of-wedlock birth shares overstate family fragility. For this reason, I treat the OECD panel as a first pass and rely on the World sample for the primary measurement.

The *World sample* addresses the OECD measurement caveat directly. It combines a global panel of children's living arrangements from Brenøe and Wasserman (2025), based on harmonized IPUMS census data from 95 countries and measuring whether children coreside with both biological parents, with UN fertility data and the female labor income share from the World Inequality Database (Chancel et al., 2022). The female labor income share is a more comprehensive measure of gender inequality than the earnings gap among workers because it incorporates participation differences. The World sample is therefore the workhorse dataset for the central empirical claim.

The two *U.S. samples* serve as a within-country check, holding the institutional environment roughly fixed. Cross-state variation in 2023 tests whether the trade-off survives when culture, law, and macroeconomic conditions are broadly shared. The 741 commuting zones for the year 2000 provide a finer geographic cut, and the corresponding results are reported in Appendix A.5.

Variable construction

For every sample, I split each of the three outcomes at its sample-specific median. "High fertility" (F) exceeds the median total fertility rate, "dual parenthood" (M) exceeds the median share of children with both parents (equivalently, falls below the median share born out of wedlock), and "gender equality" (G) falls below the median earnings gap (equivalently, exceeds the median female labor income share). Using sample medians avoids arbitrary thresholds and yields balanced categories. The specific cutoffs, which vary across samples because the underlying variables differ, appear in Table 2.

Sample	High Fertility (F)	Dual Parenthood (M)	Gender Equality (G)
OECD	TFR > 1.69	OOW $< 31.4\%$	Gap $< 17.2\%$
World	TFR > 2.81	Both parents $> 75.9\%$	Female share $> 32.3\%$
U.S. States	Births/1000 > 56.9	Single mom $< 17\%$	Gap $< 25.6\%$
U.S. CZ	CBR > 12.9	Single mom $< 19.7\%$	Ratio < 1.80

Table 2. Median Cutoffs for Binary Indicators

Notes: TFR denotes the total fertility rate (children per woman). OOW denotes the out-of-wedlock birth share. CBR denotes the crude birth rate (births per 1,000 population). Gap is defined as (male median earnings − female median earnings) divided by male median earnings. Ratio is defined as male median earnings divided by female median earnings.

A primer on mutual information

A Venn diagram reveals how often all three outcomes overlap, but it cannot tell us whether the scarcity of joint attainment is simply the product of two pairwise trade-offs already documented in the literature. To answer that question, I turn to information theory.

Mutual information $I(X;Y)$ measures how much knowing the value of one binary variable reduces uncertainty about another. It is zero when the variables are independent and grows as they become more tightly linked. For three variables (X, Y, Z), the relevant object is *interaction information*, defined as the difference between the pairwise mutual information $I(X;Y)$ and the conditional mutual information $I(X;Y \mid Z)$. Intuitively, interaction information compares how informative X is about Y on its own with how informative X is about Y once Z is known.

Three cases matter. If conditioning on Z leaves the X–Y relationship unchanged, interaction information is zero and the three variables decompose into independent pairs. If conditioning on Z *weakens* the X–Y relationship, interaction information is positive and the three variables are *redundant*, meaning that Z already contains some of what X tells us about Y. If conditioning on Z *strengthens* the X–Y relationship, interaction information is negative and the three variables exhibit *synergy*, meaning that they jointly carry information that none of the three pairs carries alone.

The distinction between redundancy and synergy is consequential for this paper's central question. If the empirical trade-off were merely redundant, a pairwise analysis would suffice, and existing work by Erosa et al. (2016), Voena (2015), and others would already tell the whole story. If the trade-off is synergistic, by contrast, the joint structure cannot be recovered from any combination of pairwise results. As I show next, the data deliver synergy.

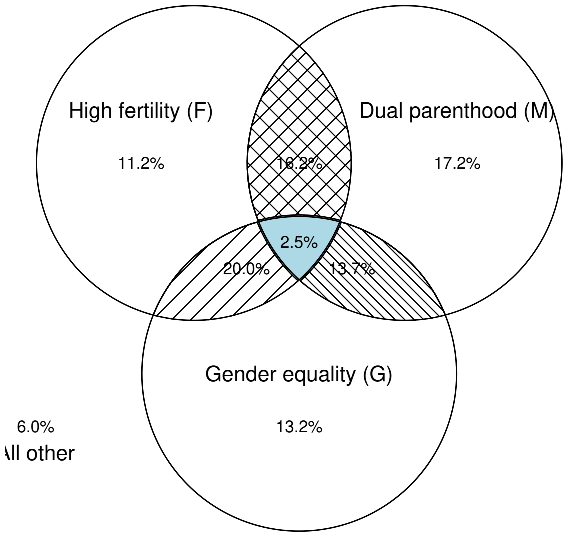
2.2 Results

The three-way trade-off is rare

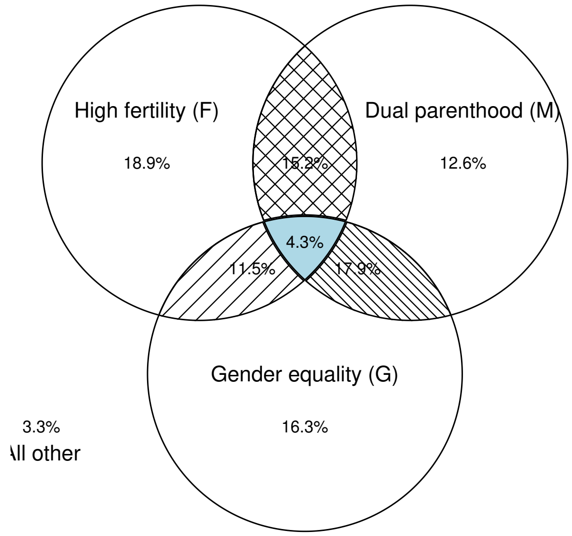
Figure 1 presents Venn diagrams for the three main samples. A uniform pattern emerges. The share of observations achieving the trinity F+M+G lies far below the 12.5 percent independence benchmark.

In the OECD panel, only 2.5 percent of country-year observations attain the trinity, one-fifth of the independence benchmark. Bootstrap inference with 5,000 replications yields a z-statistic of -16.51 with $p < 0.001$. The World sample, which uses the preferred measure of children's coresidence with both biological parents, shows 4.3 percent ($z = -12.40$, $p < 0.001$). The U.S. cross-state analysis, which holds institutions roughly constant, still shows only 5.9 percent ($z = -1.90$, $p = 0.029$). Across U.S. commuting zones, the share is 7.6 percent (Appendix A.5).

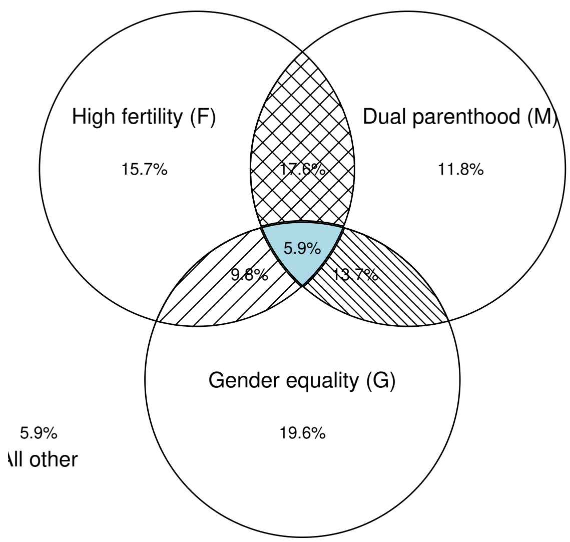
A revealing negative result is that no *pairwise* combination consistently falls below its corresponding independence benchmark of 25 percent at conventional significance levels. The tension surfaces only when all three outcomes are considered simultaneously. This is the first hint that a pairwise view misses something.



(a) OECD ($N = 721$)



(b) World ($N = 1{,}130$)



(c) U.S. States ($N = 51$)

Figure 1. Three-Way Trade-Off Across Samples

Notes: Venn diagrams showing the joint distribution of high fertility (F), dual parenthood (M), and gender equality (G), each defined at sample-specific medians (see Table 2). Under statistical independence, the central overlap (F+M+G) would contain 12.5 percent of observations. Results for U.S. commuting zones are reported in Appendix A.5.

The trade-off is synergistic, not redundant

A brief primer on mutual information. Before presenting the results, it is useful to introduce the tools. *Mutual information* is a measure from information theory that quantifies how much knowing one variable reduces uncertainty about another. It is measured in *bits*:

a value of zero means the two variables are statistically independent, and larger values mean stronger dependence. Unlike a correlation coefficient, mutual information captures dependencies of *any* form, linear or not, and makes no distributional assumptions.

When three variables are involved, a natural question is whether their joint behavior can be reduced to a collection of pairwise relationships or whether something genuinely three-way is at work. *Interaction information* answers this question. Intuitively, it compares the dependence between two variables before and after conditioning on a third:

- A **positive** value indicates *redundancy*: the three variables carry overlapping information, so conditioning on one weakens the apparent link between the other two.
- A **negative** value indicates *synergy*: conditioning on the third variable *strengthens* the relationship between the other two. The joint distribution cannot be reconstructed from pairwise pieces alone; the three-way structure is irreducible.

A related quantity, *total correlation*, sums up all the dependence (pairwise and higher-order) in the joint distribution and serves as a benchmark for the overall strength of association.

Results. Table 3 reports these quantities for each of the three samples. The column of direct interest is *Interaction*; a negative value indicates synergy.

	Pairwise MI			Higher-Order		
Sample	$I(F;M)$	$I(F;G)$	$I(M;G)$	Total Corr.	Interaction	Interpretation
OECD	0.043	0.006	0.081	0.165	-0.035^{***}	Synergy
World	0.034	0.099	0.009	0.169	-0.026^{***}	Synergy
U.S. States	0.000	0.082	0.023	0.112	-0.007	Synergy

Table 3. Mutual Information Analysis Across Samples

Notes: All values are in bits. F denotes high fertility, M denotes dual parenthood, and G denotes gender income equality. Pairwise MI denotes pairwise mutual information. Total correlation is the sum of pairwise MI minus a joint entropy adjustment, and summarizes the total dependence among the three variables. Interaction information measures the higher-order component: negative values indicate synergy, meaning that conditioning on any one of the three variables *increases* the dependence between the other two. Bootstrap inference uses 5,000 replications: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. See Appendix Table A.1 for detailed results, including conditional mutual information, subsample analyses, and U.S. commuting zone results.

Interaction information is negative in all three samples and statistically significant in the two cross-country panels. The OECD sample illustrates the mechanism most transparently. The unconditional mutual information between fertility and dual parenthood is $I(F;M) = 0.043$ bits. A standard confounding story would predict that this link should *shrink* once we account for gender equality: if G were simply driving both F and M, then comparing countries *within* the same gender-equality regime should leave little residual association. The data show the opposite. The conditional mutual information rises to $I(F; M \mid G) = 0.078$ bits—nearly double the unconditional value. Pooling gender-equal and non-gender-equal countries blurs the fertility–dual-parenthood relationship because the two groups follow distinct patterns; separating them makes the link sharper, not weaker. The World sample exhibits the same phenomenon for a different pair: $I(F; G \mid M) = 0.125$ bits exceeds $I(F; G) = 0.099$ bits. The U.S. cross-state sample points in the same direction but is not statistically significant, consistent with the compressed variation observed within a single country.

Why this matters for the literature. The finding of synergy has a concrete methodological implication: the three outcomes—fertility, dual parenthood, and gender income equality—genuinely belong in the same frame, and studying them pairwise leaves structure on the table. Existing work has, understandably, focused on two-variable slices: fertility and earnings in Erosa et al. (2016), marriage and fertility in Regalia and Rios-Rull (2001), and marriage and earnings in Stevenson and Wolfers (2007). Each of these studies captures a projection of a distribution whose essential character is three-dimensional. A synergistic joint structure cannot, by construction, be recovered from its two-dimensional marginals; the component of the dependence that lives in the three-way interaction is invisible to any pairwise analysis. This is what provides the statistical rationale for treating the three outcomes as a unified object.

Transition paths and geographic distribution

Figure 2 traces OECD countries' movements across the eight categories over time. A dominant pattern is the departure from F+M toward M+G or G alone: countries typically secure gender equality by giving up fertility, and often give up dual parenthood in the process. Appendix Figure A.4 shows analogous paths for the World sample.

Figure 3 maps joint attainment across U.S. states. The three states that achieve the trinity, namely Alaska, Hawaii, and Minnesota, are geographically and demographically diverse. Minnesota in particular stands out for its combination of high female labor force participation with relatively high fertility and intact families, a pattern consistent with a state-level institutional environment that reduces the effective time cost of children. I return to this theme in the theoretical analysis.

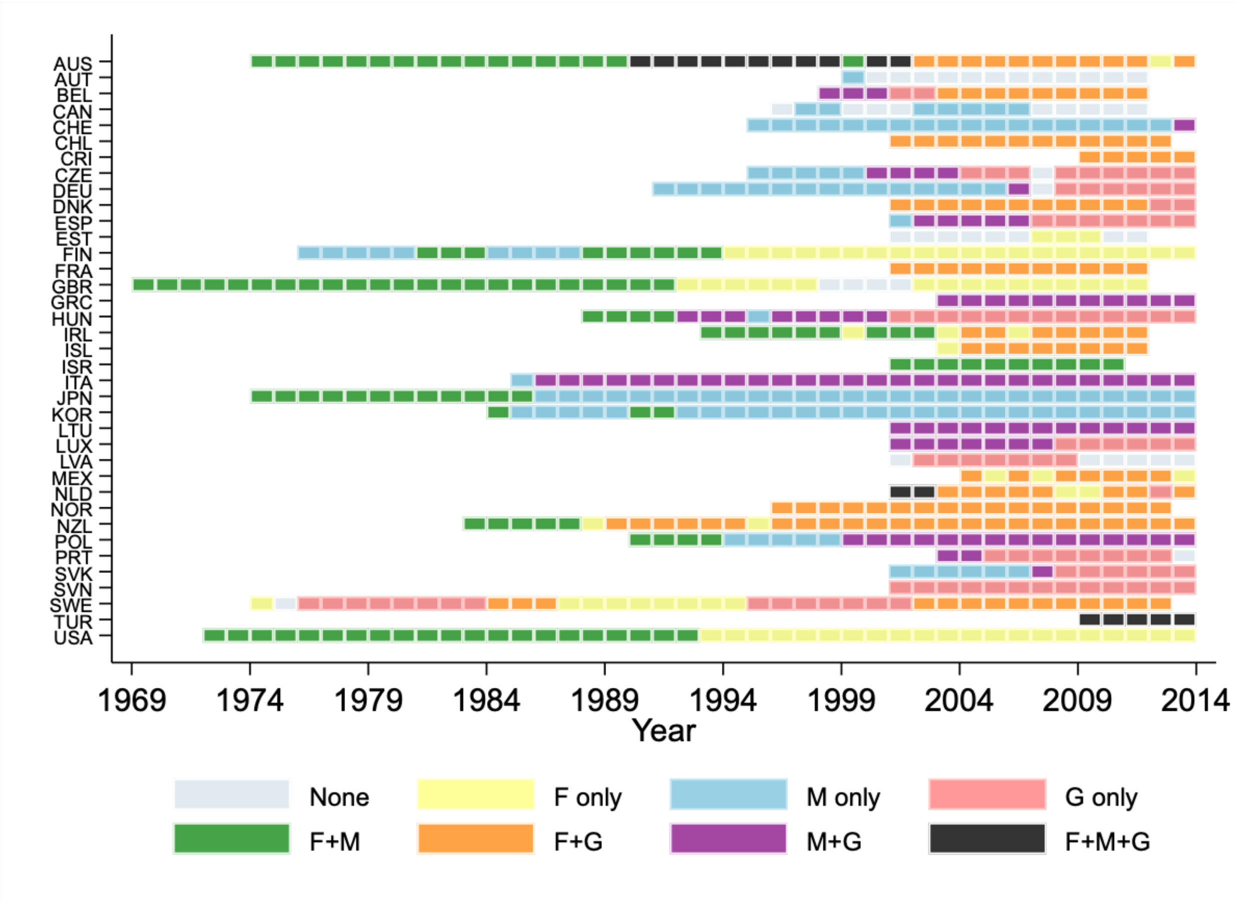


Figure 2. Transition Paths by Country (OECD Sample)

Notes: Temporal transitions in the three categories for each country (F = high fertility, M = dual parenthood, G = gender earnings equality). Countries are labeled by three-letter ISO codes. Data cover 37 OECD countries from 1970 to 2014. See Appendix Figure A.4 for the corresponding figure for the World sample.

Robustness

To probe whether the trade-off reflects a single income or education regime, I split each sample at the median of GDP per capita and at the median of average years of schooling from Barro and Lee (2013), and I redefine the three indicators at subsample medians. Appendix Figures A.2 and A.3 show Venn diagrams by subsample, and Appendix Table A.1 reports the corresponding mutual information analysis. The trade-off and the synergy result persist in most subsamples, so they are not an artifact of pooling countries at different stages of development.

2.3 Interpretation

Two findings emerge. First, simultaneously achieving high fertility, widespread dual parenthood, and gender income equality is rare. The joint share ranges from 2.5 to 7.6 percent of

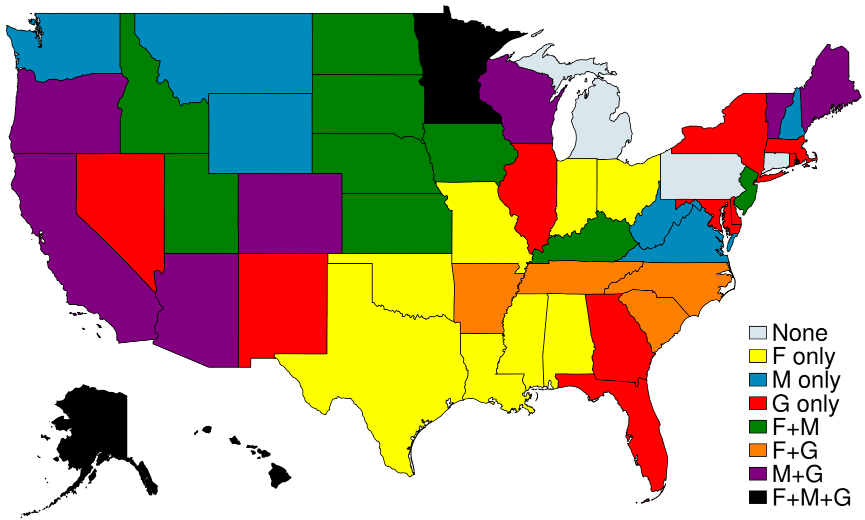


Figure 3. Joint Attainment by U.S. State (2023)

Notes: The map indicates which outcomes (F = high fertility, M = dual parenthood, G = gender earnings equality) each state achieves based on median thresholds. See Appendix Figure A.6 for the corresponding map at the commuting zone level.

observations, consistently below the 12.5 percent independence benchmark, and the shortfall is statistically significant. Second, the three outcomes exhibit synergy, so that conditioning on any one of them strengthens the dependence between the other two.

The synergy finding has a practical implication that reaches beyond the theoretical literature. Policies targeted at any one outcome will operate in an environment where the other two respond. A complete analysis of family and gender policy therefore requires a framework in which fertility, marriage, and the gender income gap are determined jointly. Section 3 provides such a framework.

3 Model

This section develops an equilibrium marriage market model in which fertility, marriage, and gender income equality emerge jointly. The model is stylized by design. Its purpose is to clarify the equilibrium logic of the three-way trade-off.

3.1 Setup

Individuals are indexed by gender $g \in \{♂, ♀\}$, with each gender comprising half the population. Utility depends on consumption c and fertility n through a constant-elasticity-of-substitution aggregator:

$$u(c,n) = \Big((1-\beta)\cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}}\Big)^{\frac{\rho}{\rho-1}}, \qquad \beta \in (0,1). \tag{1}$$

Following Jones and Schoonbroodt (2010) and Carlos Córdoba and Ripoll (2019), I assume $\rho > 1$ so that consumption and fertility are substitutes and utility is well-defined for childless individuals, with $u(c,0) = (1-\beta)^{\rho/(\rho-1)}c$.

To focus on between-gender disparities, wages are uniform within each gender, with $w^{♂}$ for men and $w^{♀}$ for women.[1] The gender wage gap is defined as

$$\Gamma^w := \frac{w^{♂}}{w^{♀}}, \tag{2}$$

so a higher Γ^w indicates a larger disparity. Throughout, Γ^w is treated as an exogenous input.[2]

Single individuals

Single men supply one unit of labor inelastically, consume their labor income, and remain childless:

$$V^{♂,s} = u(w^{♂}, 0). \tag{3}$$

Single women can bear children without support from the father.[3] They choose consumption $c^{♀,s}$, labor supply l^s, and fertility n^s to maximize

$$V^{♀,s} = \max_{c^{♀,s},l^s,n^s} \quad u(c^{♀,s}, n^s) \tag{4}$$

subject to

$$c^{♀,s} = w^{♀}l^s, \qquad l^s = 1 - \chi n^s,$$

[1] Abstracting from within-gender heterogeneity in wages or education shuts down assortative matching. Incorporating such heterogeneity would require assortative-matching machinery (Chiappori, 2017), which is beyond the scope of this paper. It would add richness to the within-gender distribution of outcomes but would not overturn the three-way trade-off, which operates on the between-gender margin. Empirically, Figure A.7 shows that the spousal earnings gap among newly-wed U.S. couples without children is 20 to 30 percent, well above the 10 percent gap for young workers, so assortative mating does not undo the within-couple gender gap.

[2] In a richer life-cycle model such as Erosa et al. (2016) or Blundell et al. (2016), reductions in female labor supply would feed back into a larger Γ^w through depreciated human capital. Such feedback would *strengthen* the three-way trade-off, because the channel linking fertility to the gender income gap would operate not only through contemporaneous labor supply but also through human capital accumulation. The baseline treatment here is therefore conservative.

[3] Child support enforcement exists in practice but is limited. According to the Annie E. Casey Foundation, using 2020–2022 Current Population Survey data, only 23 percent of U.S. female-headed families with children under 18 received any child support in the prior year.

where $\chi > 0$ is the time cost per child. Fertility $n^s \in \mathbb{R}_+$ is treated as continuous, following standard models of female labor supply and fertility (De La Croix and Doepke, 2003).

The setup abstracts from two margins that have received careful treatment elsewhere. It does not include childcare marketization (Bar et al., 2018) in which households purchase childcare as a substitute for parental time, and it does not model the quantity-quality trade-off of Becker and Lewis (1973) explicitly. Section 3.4 nonetheless considers policies that reduce the time cost χ directly, and because consumption c and fertility n are substitutes in both preferences and budget, the reduced-form mechanism of the quantity-quality trade-off is preserved.

Married individuals

In marriage, husbands supply one unit of labor inelastically and transfer a share α of their income to their wives.[4] This transfer captures the economic function of marriage in supporting child-rearing. Individuals take α as given; it is determined in equilibrium, as detailed in Section 3.2. The support is $\alpha \in [-w^{♂}/w^{♀}, 1]$ so that both consumptions are positive. Negative values of α (transfers from wives to husbands) are permitted in principle.

Husbands consume their remaining income and derive utility from shared fertility n^m and a psychic benefit of marriage λ:

$$V^{♂,m} = \max_{n^m} \quad u\Big((1-\alpha)w^{♂}, n^m\Big) \cdot \lambda. \tag{5}$$

Wives choose consumption $c^{♀,m}$, labor supply l^m, and fertility n^m to maximize

$$V^{♀,m} = \max_{c^{♀,m}, l^m, n^m} \quad u(c^{♀,m}, n^m) \cdot \lambda \tag{6}$$

subject to

$$c^{♀,m} = \alpha w^{♂} + w^{♀} l^m, \qquad l^m = 1 - \chi n^m.$$

Husbands' transfers generate a pure income effect for wives, so both consumption and fertility rise with α. Higher female wages, by contrast, induce a substitution effect; since $\rho > 1$, rising $w^{♀}$ reduces desired fertility, which is the mechanism central to Greenwood

[4] Modeling the transfer as a fixed share of male income, rather than as a contract contingent on realized fertility, is a deliberate simplification. It avoids the time-consistency issues that would arise if spouses could renegotiate the transfer after children are born, and it keeps the equilibrium object one-dimensional in α. Allowing for limited commitment would attenuate, but not reverse, the quantitative results: once children are born, a wife's outside option deteriorates because single motherhood is costlier than singlehood, so husbands could renegotiate α downward ex post. Anticipating this hold-up, wives would choose lower marital fertility, dampening the fertility response to changes in Γ^w, but the directional mechanism—a wider wage gap raising the value of marriage and hence marital fertility—is preserved.

et al. (2005a). Higher male wages increase fertility via larger transfers, consistent with the evidence in Jakobsen et al. (2022).

Marriage market

At the start of each period, each individual draws an idiosyncratic taste shock ε for marriage relative to singlehood, distributed Fréchet with scale θ. By McFadden (1973), the gender-specific marriage rates are

$$\mathcal{M}^{♂} = \frac{1}{1 + (V^{♂,s}/V^{♂,m})^{\theta}}, \qquad \mathcal{M}^{♀} = \frac{1}{1 + (V^{♀,s}/V^{♀,m})^{\theta}}. \tag{7}$$

Equilibrium in the marriage market requires

$$\mathcal{M}^{♂} = \mathcal{M}^{♀} = \mathcal{M}, \tag{8}$$

with the pair (α, n^m) playing the role of market-clearing prices.[5]

This is where the model departs from the scalar-transfer tradition of Choo and Siow (2006). In that tradition, a single transferable-utility object clears the market. Here, because marriage bundles economic support with shared fertility, the price is two-dimensional: α transfers resources, and n^m transfers children. Both must adjust in equilibrium, and the interaction between them is what generates the three-way trade-off.

Aggregate variables

Aggregate fertility is the population-weighted average of marital and non-marital fertility:

$$n = \mathcal{M} \cdot n^m + (1 - \mathcal{M}) \cdot n^s. \tag{9}$$

The share of children living with both parents is

$$\mathcal{D} = \frac{\mathcal{M} \cdot n^m}{n}. \tag{10}$$

Average female labor supply simplifies to

$$l^{♀} = \mathcal{M} \cdot l^m + (1 - \mathcal{M}) \cdot l^s = 1 - \chi n. \tag{11}$$

[5]The equilibrium concept is competitive in the spirit of Choo and Siow (2006): each agent is atomistic and takes (α, n^m) as given. What is non-standard relative to that tradition is that the price is two-dimensional, reflecting the joint nature of marriage as both an income-sharing and a fertility-sharing arrangement. Lemma 1 below establishes uniqueness.

Finally, labor income is $y^{\male} = w^{\male}$ and $y^{\female} = w^{\female} l^{\female}$, so the gender income gap is

$$\Gamma^y = \frac{y^{\male}}{y^{\female}} = \frac{\Gamma^w}{l^{\female}}. \tag{12}$$

Equation (12) captures a key point. The gender income gap inherits the wage gap, but the gap is amplified whenever women reduce their labor supply, which in equilibrium happens whenever fertility rises.

3.2 Equilibrium Characterization

Before solving the model, I state the equilibrium definition formally.

Definition 1 (Competitive equilibrium)**.** *Given primitives $(\beta, \rho, \chi, \lambda, \theta)$ and wages $(w^{\male}, w^{\female})$, a competitive equilibrium consists of single women's choices $(c^{\female,s}, l^s, n^s)$, married couples' choices $(c^{\female,m}, l^m, n^m)$, a transfer share α, gender-specific marriage rates $(\mathcal{M}^{\male}, \mathcal{M}^{\female})$, and an aggregate marriage rate $\mathcal{M}$ such that:*

1. *Single women optimize. Given $w^{\female}$, the tuple $(c^{\female,s}, l^s, n^s)$ solves the single-woman problem with $c^{\female,s} = w^{\female} l^s$ and $l^s = 1 - \chi n^s$. Single men are childless and consume $w^{\male}$.*

2. *Wives optimize. Taking $(\alpha, w^{\male}, w^{\female})$ as given, $(c^{\female,m}, l^m, n^m)$ solves the married-woman problem with $c^{\female,m} = \alpha w^{\male} + w^{\female} l^m$ and $l^m = 1 - \chi n^m$. Because fertility in marriage requires mutual consent, n^m is pinned down by the wife's first-order condition.*

3. *Husbands participate. Taking (α, n^m) as given, the husband's value $V^{\male,m}$ in (7) is evaluated at n^m chosen by the wife, and α satisfies the husband's participation condition implied by marriage-market clearing.*

4. *Marriage decisions are individually rational. Given idiosyncratic Fréchet taste shocks, the gender-specific marriage rates $\mathcal{M}^{\male}$ and $\mathcal{M}^{\female}$ are given by (7).*

5. *Marriage market clears. The two-dimensional price (α, n^m) adjusts so that*

$$\mathcal{M}^{\male} = \mathcal{M}^{\female} = \mathcal{M}.$$

I now characterize this equilibrium in three steps.

First, for a given marriage rate $\mathcal{M}$, the husband's first-order condition and the marriage market equation (7) together imply α as a function of marital fertility n^m:

$$\alpha = 1 - \left[\left(\left(\frac{\mathcal{M}}{1-\mathcal{M}} \right)^{\frac{1}{\theta}} \cdot \frac{1}{\lambda} \right)^{\frac{\rho-1}{\rho}} - \frac{\beta}{1-\beta} \cdot \left(\frac{n^m}{w^{♂}} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}. \tag{13}$$

Since $\rho > 1$, the function $\alpha(n^m)$ is strictly increasing and concave, and it equals zero when $n^m = 0$.[6] The intuition is that husbands require larger transfers to sustain higher marital fertility. When marital fertility is zero, all gains from marriage accrue through the psychic benefit λ, and the marginal man requires no pecuniary compensation. For fixed n^m, α decreases in $w^{♂}$, because a given transfer share represents more resources when male income is higher.

Second, the first-order condition for married women determines n^m as a function of α:

$$n^m \cdot \left[\left(\frac{(1-\beta)\chi}{\beta} \right)^{\rho} \cdot (w^{♀})^{\rho-1} + \chi \right] = 1 + \alpha \Gamma^w. \tag{14}$$

Marital fertility is linear and increasing in α. Even when $\alpha = 0$, marital fertility is strictly positive, and comparing to the single-women's first-order condition shows that it equals n^s in that case. Higher female wages shift $n^m(\alpha)$ downward because the substitution effect dominates the income effect when $\rho > 1$.

The two relations (13) and (14) jointly pin down (α, n^m). To guarantee that this system has a unique interior solution, I impose the following mild condition on primitives.

Assumption 1. *The psychic benefit of marriage λ satisfies*

$$\lambda \geq \left(\frac{1-\beta}{\beta} \right)^{\frac{1}{\rho-1}} \cdot \frac{1}{w^{♂}\chi} \cdot \left[\left(\frac{(1-\beta)\chi}{\beta} \right)^{\rho} (w^{♀})^{\rho-1} + \chi \right]. \tag{15}$$

This condition ensures that the marginal man in equilibrium does not require compensation ($\alpha \geq 0$) even if his wife remains childless.

The bound depends only on primitives $(\beta, \rho, \chi, w^{♂}, w^{♀})$ and can be verified. In the Mexican calibration, the estimated $\lambda = 1.03$ satisfies the condition in every year.

Lemma 1. *Under Assumption 1, for any wage pair $\{w^{♂}, w^{♀}\}$, there exists a unique fixed point $(\alpha, n^m) \in (0,1) \times \mathbb{R}_{++}$ that clears the marriage market.*

[6] The property that $\alpha = 0$ when $n^m = 0$ is a consequence of the equilibrium rather than an imposed assumption. When marital fertility is zero, all gains from marriage accrue through the psychic benefit λ, and Equation (13) then requires the marginal man to be exactly indifferent between marriage and singlehood, which pins α at zero. Assumption 1 below ensures that the equilibrium α remains non-negative for any $n^m \geq 0$.

Proof: See Appendix C.2.

Lemma 2. *The gains from marriage for women are* $V^{♀,m}/V^{♀,s} = \lambda \cdot (1 + \alpha\Gamma^w)$.

Proof: See Appendix C.3.

Lemma 2 summarizes the intuition compactly. Women's gains from marriage are the product of the psychic benefit λ and an economic component $1 + \alpha\Gamma^w$. The economic component itself is the product of the share of male income transferred to the wife and the gender wage gap. This result revives, in a two-dimensional price setting, the classic Becker (1973) observation that marriage becomes more attractive to women when the wage gap is wider. Combined with Equation (31) in the appendix, Lemma 2 also implies that the dual-parenthood share $\mathcal{D}$ is strictly increasing in the marriage rate $\mathcal{M}$, so the two objects can be used interchangeably in what follows.

3.3 The Three-Way Trade-Off

The equilibrium conditions reduce to three equations:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha\Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha\Gamma^w))^\theta}, \tag{16}$$

$$\Gamma^y = \frac{\Gamma^w}{l^{♀}}, \tag{17}$$

$$l^{♀} = 1 - \chi n. \tag{18}$$

These three equations expose the trade-off transparently. A high marriage rate requires a large economic surplus $\alpha\Gamma^w$, which in turn requires either a large transfer share or a large wage gap. But a large wage gap widens Γ^y unless female labor supply is high, and high female labor supply requires low fertility. The three possible pairs of outcomes therefore run into a common obstacle:

(1) *High fertility and high dual parenthood.* High n implies low $l^{♀}$ through (18). Sustaining high $\mathcal{M}$ then requires a sizable Γ^w through (16), which widens Γ^y through (17).

(2) *High fertility and gender income equality.* High n again implies low $l^{♀}$. Low Γ^y then forces a very small Γ^w through (17), and since α is bounded above, the economic surplus from marriage $\alpha\Gamma^w$ shrinks and $\mathcal{M}$ falls through (16).[7]

(3) *High dual parenthood and gender income equality.* High $\mathcal{M}$ again requires a sizable Γ^w through (16). Offsetting the wage gap to deliver low Γ^y requires high $l^{♀}$ through (17), which pushes fertility down through (18).

[7] In practice, the upper bound on α may lie well below one due to norms or subsistence constraints.

The trade-off is the equilibrium counterpart of the empirical synergy documented in Section 2. The three outcomes are tied together by a common set of equilibrium relations rather than by three separate pairwise mechanisms.

3.4 Policy Remedies

Policymakers often target fertility, marriage, and gender equality as if they were independent objectives. In the model, they are not. I evaluate three standard policy instruments in turn and then introduce a fourth that can loosen the trade-off.

Pro-marriage policies

Joint taxation, marriage bonuses, and similar pro-marriage measures are motivated both by cultural considerations and by evidence that family structure shapes the intergenerational transmission of skill (Kearney, 2023; Autor et al., 2019). In the model, they correspond to an increase in the psychic benefit of marriage λ.

Proposition 1. *An increase in the psychic benefit of marriage λ raises aggregate fertility n and the marriage rate $\mathcal{M}$ but widens the gender income gap Γ^y.*

Proof: See Appendix C.4.

The logic is compositional. A higher λ lifts $\mathcal{M}$ without directly changing n^m. Because $n^m > n^s$, average fertility rises mechanically. Higher fertility then reduces female labor supply and widens the income gap.

Gender equality policies

Anti-discrimination laws and pay equity measures aim to reduce labor market misallocation (Hsieh et al., 2019). In the model, they raise $w^{♀}$ while holding $w^{♂}$ fixed.

Proposition 2. *An increase in the female wage $w^{♀}$ (holding $w^{♂}$ fixed) reduces aggregate fertility n, lowers the marriage rate $\mathcal{M}$, and narrows the gender income gap Γ^y.*

Proof: See Appendix C.5.

Higher female wages push down both n^s and n^m through substitution. Lower n^m reduces the equilibrium transfer α, and the shrinking economic surplus $\alpha\Gamma^w$ lowers the marriage rate. Female labor supply rises, and combined with the smaller wage gap, the gender income gap narrows.

Uniform reductions in childcare costs

Baby bonuses, public childcare, and similar subsidies aim to counter population aging (Bongaarts, 2004; Jones, 2022). In the baseline model, they reduce the time cost of children χ.

Proposition 3. *A reduction in the child-rearing cost χ increases aggregate fertility n, raises the marriage rate $\mathcal{M}$, and widens the gender income gap Γ^y.*

Proof: See Appendix C.6.

Lower χ raises both n^s and n^m. The resulting increase in n^m supports a higher equilibrium α and a higher marriage rate, which amplifies the effect on aggregate fertility. Female labor supply actually *falls*: the income effect through higher fertility dominates the direct effect of the lower time cost, and the gender income gap widens.

Three instruments, three incomplete solutions

Table 4 collects the comparative statics.

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{♀}$)	–	–	– (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

Table 4. Summary of Policy Effects

Notes: Signs indicate the direction of change in each outcome variable in response to the indicated policy shift.

None of the three instruments can resolve the three-way trade-off. Pro-marriage and pro-fertility policies raise fertility and marriage at the cost of gender equality. Gender equality policies narrow the income gap but depress fertility and marriage. The trade-off is binding.

There is a related fiscal observation. Because the policy cost of subsidizing childcare rises with $w^{♀}$, standard pro-fertility policies become progressively more expensive as gender wage gaps close. The government is effectively subsidizing a non-market sector with no productivity growth, and as the opportunity cost of women's time rises, so does the fiscal bill.

3.5 Redistributing childcare between spouses

The baseline model assumes that only wives bear the time cost of children. Time-use evidence across countries confirms that mothers spend substantially more time on childcare than fathers (Aguiar and Hurst, 2007; Kleven et al., 2019), but the division of childcare labor is neither fixed nor immutable. It responds to policies such as reserved paternity leave quotas (Ahn and Mira, 2002; Farré and González, 2019). To study what changes when the allocation of childcare is itself a policy lever, I extend the model to allow gender-differentiated childcare costs.

Let $\chi^{♀}$ and $\chi^{♂}$ denote the time costs per child borne by wives and husbands within marriage, with $\chi^{♀} > \chi^{♂}$. Single mothers continue to bear the full cost χ. The wife's budget becomes

$$c^{♀,m} = \alpha w^{♂}(1 - \chi^{♂} n^m) + w^{♀}(1 - \chi^{♀} n^m), \tag{19}$$

and the husband's budget becomes

$$c^{♂,m} = (1 - \alpha) w^{♂}(1 - \chi^{♂} n^m). \tag{20}$$

Consider a policy that increases $\chi^{♂}$ and decreases $\chi^{♀}$ by equal amounts $\delta > 0$, keeping the total childcare time $\chi^{♂} + \chi^{♀}$ and the single-mother cost χ fixed. Full derivations appear in Appendix D. The next proposition characterizes when redistribution raises all three outcomes.

Proposition 4. *Consider the extended model with gender-differentiated childcare costs. Suppose (i) $\chi^{♀} > \chi^{♂}$, (ii) $w^{♂} > w^{♀}$, and (iii) $\rho > 1$. Define the wife's effective childcare cost as*

$$\tilde{\chi}^{♀} \equiv \frac{\alpha w^{♂} \chi^{♂} + w^{♀} \chi^{♀}}{\alpha w^{♂} + w^{♀}},$$

the income-weighted average of the two spouses' childcare costs from the wife's perspective. Consider a redistribution ($\uparrow \chi^{♂}, \downarrow \chi^{♀}$) by equal amounts $\delta > 0$, holding $\chi^{♂} + \chi^{♀}$ and χ fixed.

(i) The wife's effective childcare cost $\tilde{\chi}^{♀}$ decreases if and only if

$$\alpha < \frac{w^{♀}}{w^{♂}}. \tag{21}$$

(ii) When condition (21) holds, marital fertility n^m increases. Single fertility n^s is unchanged.

(iii) When condition (21) holds, the wife's utility from marriage rises relative to singlehood.

(iv) The husband's utility from marriage rises if and only if

$$\frac{\beta}{1-\beta} \cdot \left(\frac{c^{\male,m}}{n^m}\right)^{\frac{1}{\rho}} > (1-\alpha) w^{\male} \cdot \left(\frac{n^m}{\partial n^m / \partial \delta} + \chi^{\male}\right), \tag{22}$$

where $\partial n^m/\partial\delta > 0$ is the fertility response to redistribution. The condition is more likely to hold when (a) the preference weight on children β is large, (b) the husband's initial childcare share $\chi^{\male}$ is small, (c) the elasticity of substitution ρ is low, and (d) the fertility response is strong.

(v) When conditions (21) and (22) both hold, both spouses prefer marriage at higher rates, so the marriage rate $\mathcal{M}$ increases.

(vi) Aggregate fertility n increases whenever marital fertility rises and the marriage rate does not decline.

Proof: See Appendix D.4.

The logic operates through three channels. First, redistributing childcare toward husbands lowers the wife's *effective* cost of children whenever condition (21) holds, that is, whenever her own earnings exceed the transfer she receives. The intuition is that redistribution shifts the time burden away from the income source that looms largest in her budget. A lower $\tilde{\chi}^{\female}$ then raises marital fertility.

Second, under the same condition, the wife's gains from marriage rise unambiguously. A key technical result in Appendix D.4 establishes that the product $n^m \cdot \Psi^m$ is strictly decreasing in $\tilde{\chi}^{\female}$ for *all* $\rho > 1$. The wife's benefit therefore does not depend on delicate restrictions on the elasticity of substitution.

Third, the husband's net gain is captured by condition (22), which compares the marginal utility from higher fertility (left-hand side) against the marginal utility loss from reduced consumption (right-hand side). The husband benefits when preferences weight children heavily (large β), when fathers start from a low childcare baseline (small $\chi^{\male}$), when consumption and children are not highly substitutable (ρ close to one), or when the fertility response is strong.

When both conditions hold, the policy is Pareto-improving within the household, marital fertility rises, and the marriage rate climbs. Aggregate fertility n increases, dual parenthood rises, and because total childcare time within marriage shifts partly onto husbands, female labor supply increases too, narrowing Γ^y. This is the one instrument, among those studied here, that can relax all three dimensions of the trade-off simultaneously. More generally, the baseline policies and Proposition 4 together yield a sharper statement of the central theoretical takeaway: the three-way trade-off binds under policies that operate on wages or on the *level* of child-rearing costs, and is relaxed only by policies that operate on the *distribution* of child-rearing time between spouses.

What happens when the conditions fail

The two conditions are not guaranteed to hold everywhere. Two natural questions arise. What if the wife's transfer exceeds her own earnings, so condition (21) fails? What if the husband's preference for children is too weak relative to his consumption loss, so condition (22) fails? Proposition 5 characterizes these cases.

Proposition 5. *Consider the redistribution policy of Proposition 4.*

(A) Condition (21) fails *($\alpha \geq w^{♀}/w^{♂}$). The wife's effective childcare cost $\tilde{\chi}^{♀}$ weakly rises, marital fertility n^m weakly falls, the wife's gains from marriage weakly fall, and the marriage rate $\mathcal{M}$ weakly falls. Aggregate fertility n weakly falls; female labor supply $l^{♀}$ weakly rises; the gender income gap Γ^y weakly narrows.*

(B) Condition (21) holds but condition (22) fails. *Marital fertility n^m strictly rises and the wife's gains from marriage strictly rise, but the husband's gains from marriage strictly fall. The equilibrium transfer α adjusts downward to restore market clearing, and the marriage rate $\mathcal{M}$, aggregate fertility n, and the gender income gap Γ^y have ambiguous responses whose signs depend on the relative strength of the wife- and husband-side effects.*

Proof: See Appendix D.5.

The two cases produce qualitatively different patterns. Case (A) describes settings with large transfers relative to female earnings, often associated with severe wage gaps or strong full-pooling norms. Redistribution there shifts the time burden onto the income source that matters most for the wife's effective price of children, so her effective cost *rises*. The result is fewer marriages and fewer children, with the gender gap narrowing only mechanically through depressed fertility. The pattern mirrors a female-wage shock (Proposition 2) rather than a Pareto-improving reallocation.

Case (B) describes high-income, low-fertility settings in which fathers already contribute substantially at baseline and the marginal value of additional children is weak. The wife-side mechanism still operates, so marital fertility rises, but the husband's participation constraint binds: husbands resist the consumption loss, the equilibrium transfer falls, and the marriage rate can decline. This pattern is consistent with mixed evidence on Nordic-style paternity quotas, which narrow within-household gender gaps reliably but produce small or negative fertility responses.

The analysis also abstracts from potential efficiency losses if fathers are initially less productive caregivers. These caveats aside, the conditions are easily checked in any calibrated environment: I show in Section 4 that both hold comfortably for Mexico throughout 1990 to 2015, and the next subsection shows that they also organize the existing quasi-experimental evidence from elsewhere.

Organizing the mixed empirical evidence

The conditions in Propositions 4 and 5 organize the evidence on reserved paternity-leave quotas, which at first glance looks contradictory across settings. Three patterns emerge, each mapping onto a distinct channel of the model.

The gender-gap channel is robust. Reserved paternity leave raises fathers' caregiving involvement and narrows the within-household division of childcare in Spain (Farré and González, 2024), Germany (Kluve and Tamm, 2013), Norway (Cools et al., 2015), and Quebec (Patnaik, 2019). In the model, this corresponds to a reduction in $\chi^{♀} - \chi^{♂}$ and a tightening of condition (21), which requires only that α be modest relative to $w^{♀}/w^{♂}$.

The fertility channel is heterogeneous, and the heterogeneity tracks condition (22). Fertility responses are negative or null in Spain (Farré and González, 2019) and Norway (Cools et al., 2015), both high-wage, low-fertility settings in which β is plausibly modest and $\chi^{♂}$ is not especially low at baseline. Positive or neutral responses appear in Quebec (Patnaik, 2019) and among German subgroups with lower baseline paternal involvement (Kluve and Tamm, 2013). The cross-setting pattern is consistent with condition (22) binding where the model predicts it should: fertility rises where fathers start from a low childcare baseline and where preferences weight children heavily.

The third channel, joint improvement across all three outcomes, has not been tested in any single reform. This reflects the selection of reforms studied rather than a failure of the prediction: reserved paternity quotas have been adopted almost exclusively in high-income, low-fertility countries, precisely the region of the parameter space where the model predicts the fertility channel to be weakest. The parameter region in which the model predicts the largest joint gains, middle-income settings with high β, low $\chi^{♂}$, and female wages that have risen enough for condition (21) to bite, is unobserved in the quasi-experimental literature.

Proposition 4 therefore generates a falsifiable prediction. In a setting resembling the Mexican calibration of Section 4, with moderate female wages, high β, and low baseline $\chi^{♂}$, a reserved paternity-leave reform should produce a positive fertility response alongside the gender-gap narrowing already documented elsewhere. Testing this prediction requires extending the empirical frontier beyond the high-income, low-fertility settings that currently dominate the evidence base.

4 Quantitative Assessment

To illustrate the framework's quantitative content, I calibrate the model to Mexico from 1990 to 2015. The choice is motivated by three considerations. Mexico offers a long, consistent, and high-quality time series. It experienced rapid GDP growth alongside a complete reversal of the gender education gap, creating clean variation in both the level and the gender composition of wages. Its demographic trajectory is broadly representative of the transitions seen across many middle-income and developed economies, as Appendix Figure A.1 illustrates.

4.1 Data and Calibration

Figure 4 plots Mexico's transition. Fertility drops from 3.5 to 2.1 children per woman. The share of children living with both parents falls from 81 to 75 percent, and the share living only with their mother climbs from 11 to 19 percent. Women's labor income share rises from 24 to 32 percent. Real GDP per capita grows from roughly \$12,000 to over \$18,000 (2017 USD). Most striking of all, the gender education gap reverses. Women aged 25 to 35 had 0.8 fewer years of schooling than men in 1990, caught up by 2010, and pulled slightly ahead by 2015.

The calibration takes time-varying wages $\{w_t^{♀}, w_t^{♂}\}$ as the sole exogenous driver and decomposes them into three structurally interpretable components:

- $A_t = w_t^{♂}$, gender-neutral productivity;
- Γ_t^h, the gender gap in human capital;
- $B_t = (w_t^{♂}/w_t^{♀})/\Gamma_t^h$, the residual gender bias in productivity after controlling for Γ_t^h.

The procedure proceeds in three steps. First, I fix $\chi = 0.075$ from De La Croix and Doepke (2003) and construct Γ_t^h from average years of schooling using the development accounting framework of Caselli (2005). Second, I infer female labor supply $l_t^{♀}$ from observed fertility via (18) and recover wages from GDP per capita and the female labor income share. Third, I jointly calibrate the four preference and marriage parameters $\{\beta, \rho, \lambda, \theta\}$ to match the observed time paths of fertility and dual parenthood.

Identification of the four parameters is intuitive. The preference weight β controls the level of fertility, ρ governs its sensitivity to productivity growth, λ pins down the average level of dual parenthood, and θ determines its responsiveness to wages. Table 5 reports the

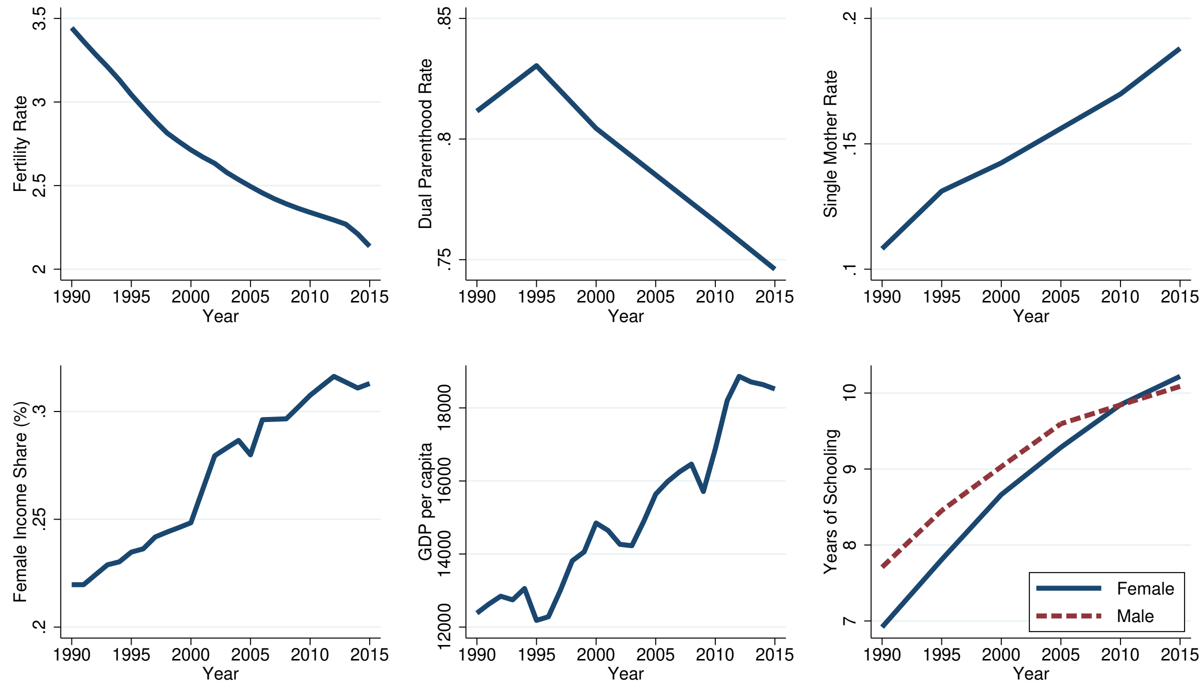


Figure 4. Mexico's Transition Path (1990–2015)

Notes: Fertility is total children per woman (United Nations). Dual parenthood is the share of children living with both biological parents; the single-mother share is the share living only with mother (Brenøe and Wasserman, 2025). The female labor income share is from the World Inequality Database. GDP per capita is in 2017 US dollars (OECD). The gender education gap is the difference in average years of schooling for ages 25–35 (Barro and Lee, 2013).

results. The estimated $\rho = 1.50$ confirms that substitution effects dominate as wages rise, and the estimated $\lambda = 1.03$ satisfies Assumption 1 in every year.

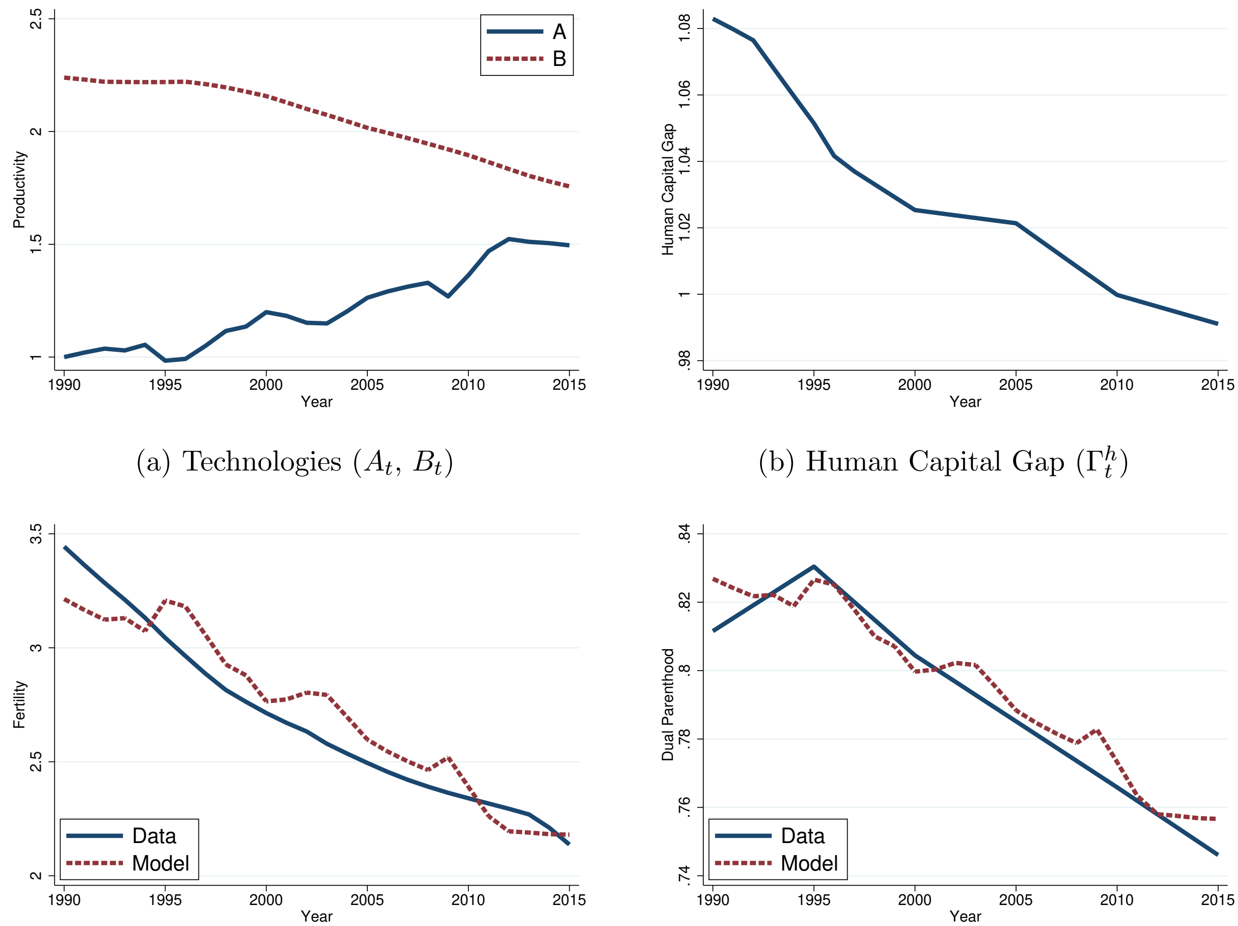
	Interpretation	Value	Target	Source
χ	time cost of children	0.075		De La Croix and Doepke (2003)
A_t	gender-neutral productivity		GDP per capita	OECD
B_t	gender-biased productivity		female labor income share	World Inequality Database
Γ_t^h	gender human capital gap		years of schooling	Barro and Lee (2013)
β	preference weight on n	0.104	fertility level	United Nations
ρ	c–n substitutability	1.50	fertility trend	United Nations
λ	psychic benefit of marriage	1.03	dual parenthood level	Brenøe and Wasserman (2025)
θ	taste shock dispersion	3.40	dual parenthood trend	Brenøe and Wasserman (2025)

Table 5. Calibration Results

Notes: Time-varying inputs (A_t, B_t, Γ_t^h) are constructed directly from data. Preference and marriage parameters minimize the distance between model and data for fertility and dual parenthood over 1990 to 2015.

Figure 5 plots the driving forces and the model fit. Gender human capital converges from

1990 onward. Gender-biased productivity B_t begins to decline in the late 1990s, and gender-neutral productivity A_t grows steadily after the 1994 Peso Crisis. With only four estimated parameters, the model closely tracks Mexico's fertility and dual-parenthood trajectories.



(a) Technologies (A_t, B_t)

(b) Human Capital Gap (Γ_t^h)

(c) Fertility

(d) Dual Parenthood

Figure 5. Model Fit: Mexico (1990–2015)

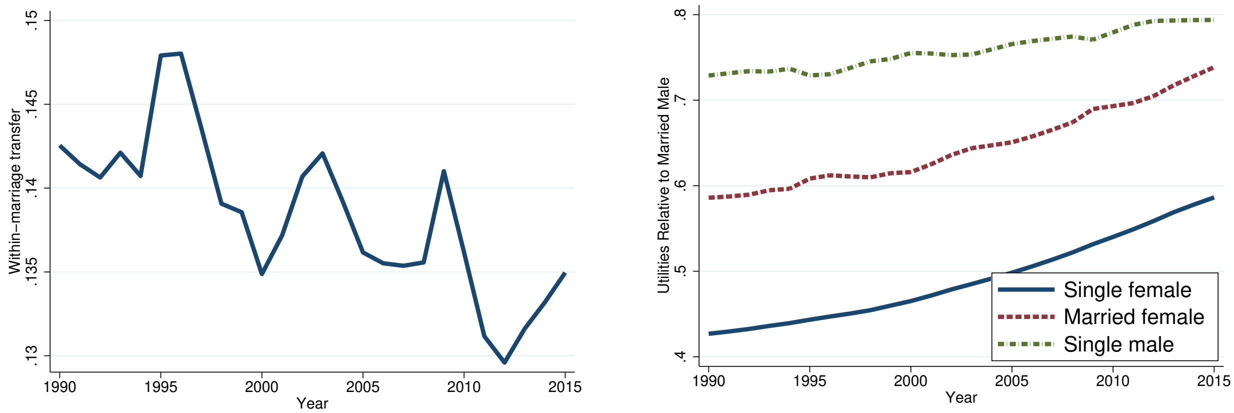
Notes: Solid lines denote data; dashed lines denote model. Fertility is total children per woman (United Nations). Dual parenthood is the share of children living with both biological parents (Brenøe and Wasserman, 2025). $A_t = w_t^{\male}$ is gender-neutral productivity (constructed from GDP per capita, OECD). $B_t = (w_t^{\male}/w_t^{\female})/\Gamma_t^h$ is gender-biased productivity (constructed from the female labor income share, World Inequality Database). Γ_t^h is the gender human capital gap (constructed from years of schooling, Barro and Lee 2013). The model is calibrated using the parameters in Table 5.

4.2 Equilibrium Outcomes

One advantage of the model is that it recovers intra-household transfers and gender-specific utility levels, objects that are difficult to observe directly. Figure 6 reports the key equilibrium quantities.

Figure 6a shows that the equilibrium transfer share α_t falls by only about one percentage point despite the sharp decline in marital fertility. The intuition is that, as the wage gap narrows, husbands must transfer a *larger share of a smaller relative income* to sustain any given marital fertility through income effects, and these two forces roughly offset in the share.

Figure 6b documents how welfare realigns across family types. Single men gain ground relative to married men as marital fertility declines. Both single and married women gain substantially relative to married men as the wage gap closes. The husband-wife welfare gap within marriage shrinks from 42 to 26 percent, even though transfers decline in absolute share. Women's rising earning power, rather than larger transfers, drives this convergence.



(a) Within-Marriage Transfers (α_t)

(b) Relative Utilities (Benchmark: $V^{♂,s}$)

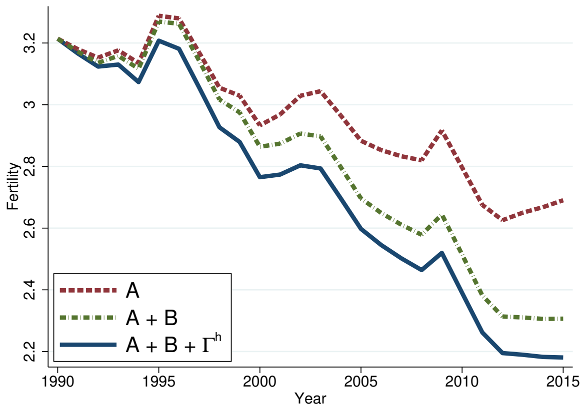
Figure 6. Model-Generated Equilibrium Outcomes

Notes: Model simulations for Mexico, 1990 to 2015, using the calibrated parameters from Table 5. Panel (a) plots the share of male income transferred to wives within marriage (α_t). Panel (b) reports the utilities of married men ($V^{♂,m}$), married women ($V^{♀,m}$), and single women ($V^{♀,s}$) relative to single men ($V^{♂,s}$).

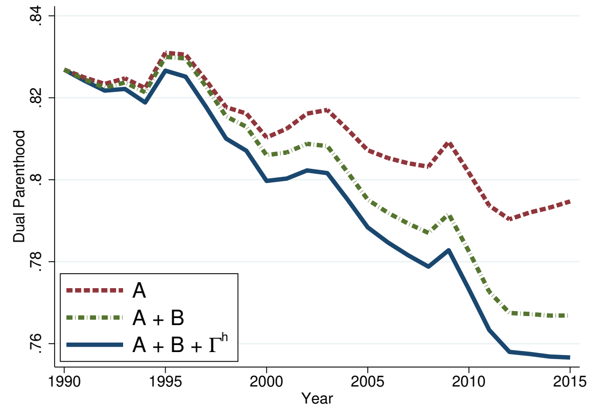
These equilibrium quantities also inform the policy analysis of Section 3.5. The equilibrium $\alpha \approx 0.10$ satisfies condition (21) throughout the sample, since $w^{♀}/w^{♂}$ ranges from about 0.45 in 1990 to 0.62 in 2015. A redistribution of childcare from wives to husbands would therefore benefit wives in Mexico's environment. Evaluating condition (22) at the calibrated equilibrium values, the left-hand side exceeds the right-hand side in every year, reflecting Mexico's high preference weight on children ($\beta = 0.104$) together with the assumption that fathers initially bear very little childcare burden. Both conditions are therefore satisfied, and redistribution would be Pareto-improving within the household in this setting.

4.3 Decomposition

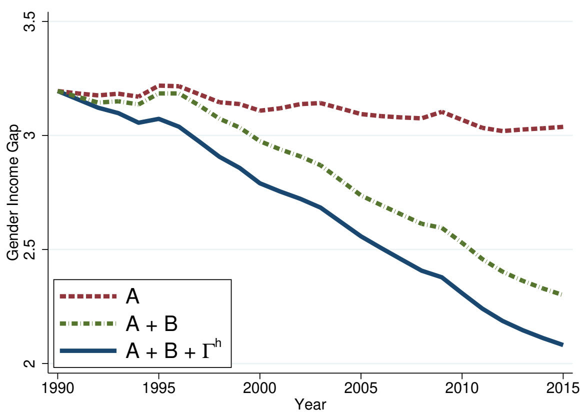
To identify the drivers of Mexico's transition, I decompose changes in fertility, dual parenthood, the gender income gap, and the spousal welfare ratio into contributions from A_t, B_t, and Γ_t^h by sequentially activating each driver.[8] Figure 7 displays the results.



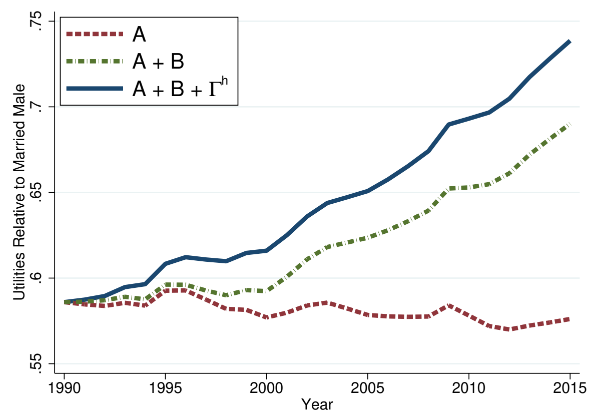
(a) Fertility



(b) Dual Parenthood



(c) Gender Income Gap



(d) Relative Utility: $V^{♀,m}/V^{♂,m}$

Figure 7. Decomposition of Mexico's Transition (1990–2015)

Notes: Solid lines denote data; dashed lines denote counterfactual paths. "A_t only" varies gender-neutral productivity while fixing B_t and Γ_t^h at their 1990 values. "$A_t + B_t$" allows both to vary while fixing Γ_t^h. "All" refers to the full model.

Gender-neutral productivity A_t alone accounts for about 50 percent of the fertility decline, operating through higher opportunity costs, and about 50 percent of the drop in dual parenthood, as lower marital fertility reduces men's gains from marriage and shifts the composition of births toward single mothers. It contributes only about 10 percent to gender income convergence, operating through higher female labor supply, and it actually widens the spousal welfare gap because lower marital fertility reduces equilibrium transfers.

Gender-biased productivity B_t explains roughly 40 percent of the fertility decline, 40 percent of the dual-parenthood decline, and the bulk of gender income and welfare convergence.

[8]Figure A.8 reports analogous decompositions for $V_t^{♀,s}/V_t^{♂,m}$ and $V_t^{♂,s}/V_t^{♂,m}$.

It operates through two channels. Absolute increases in female wages raise opportunity costs, while relative increases require larger α to sustain any given n^m, reducing marriage rates.

The reversal of Γ_t^h accounts for the remainder. Because B_t is defined as the residual wage gap net of Γ_t^h, the relative contributions of the two gender-specific drivers depend directly on how much of the observed wage-gap movement is attributable to schooling.

4.4 Prediction

I extrapolate wages linearly from the 1990 to 2015 trend and solve the model period by period through 2040. Figure 8 shows the results. Fertility falls to around 1.5 by 2040, dual parenthood drops from 75 to roughly 70 percent, and the gender income gap continues to narrow. The 2023 out-of-sample fertility prediction is 1.93, against an actual value of 1.91.

The value of the framework here is that it does not extrapolate fertility, marriage, and gender gaps independently. The three outcomes are treated as joint endogenous outcomes of gender-specific wage dynamics, which in turn are decomposed into three structurally interpretable drivers.

5 Dynamic Extension

The static framework takes the gender human capital gap as exogenous. A rich body of empirical work, however, shows that family structure shapes children's human capital differently by gender, with boys especially vulnerable to the absence of a biological father (Sommers, 2001; Bertrand and Pan, 2013; Chetty et al., 2016; Autor et al., 2019; Wasserman, 2020; Reeves, 2022; Frimmel et al., 2024). Autor et al. (2019) trace much of the black-white gender gap in outcomes to differences in single motherhood, and Autor et al. (2023) find that family structure accounts for a substantial share of gender disparities in high school performance. These patterns suggest a feedback loop worth incorporating.

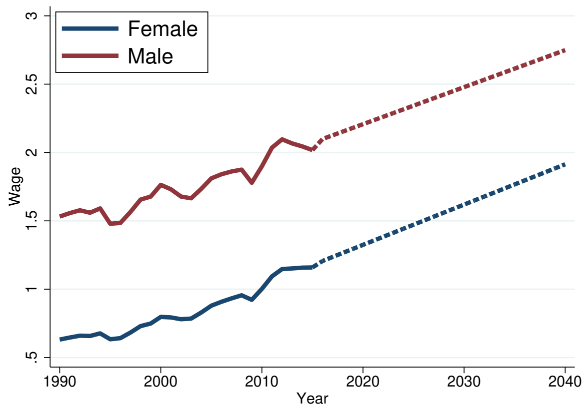
I add the simplest law of motion consistent with this evidence. The next-period human capital gap depends positively on the current marriage rate,

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t), \qquad \mathcal{H}'(\mathcal{M}_t) > 0. \tag{23}$$

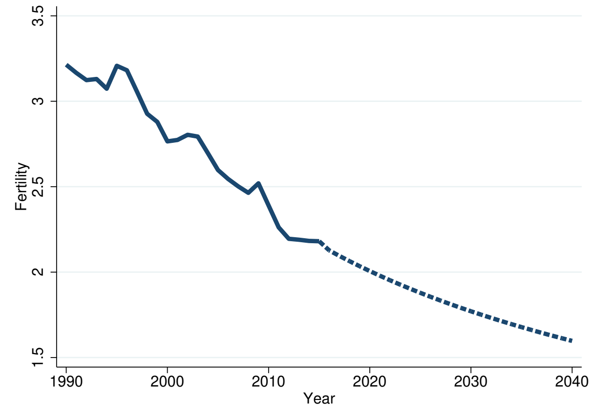
The assumption $\mathcal{H}' > 0$ captures the differential impact of single parenthood on boys, whether through same-gender role models, differential sensitivity to parental inputs, or differential exposure to peer and neighborhood environments.

Figure 9 summarizes the dynamic mechanism.

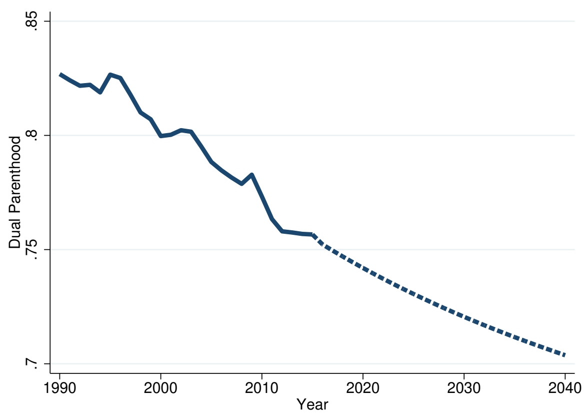
The decline unfolds through two exogenous drivers and one reinforcing feedback:



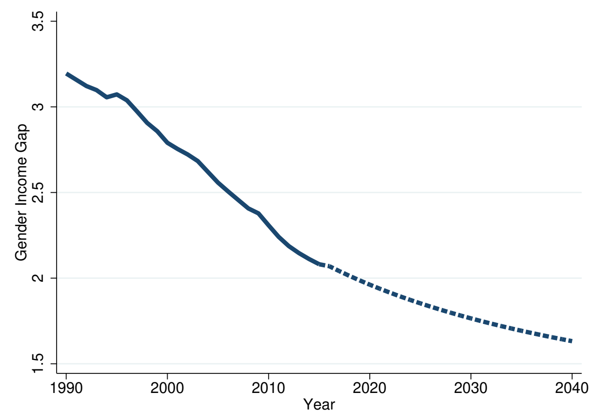
(a) Extrapolated Wages



(b) Fertility



(c) Dual Parenthood



(d) Gender Income Gap

Figure 8. Model-Based Predictions: Mexico to 2040

Notes: Dashed lines denote linear extrapolation of gender-specific wages from the 1990 to 2015 trend. Solid lines denote model predictions using the calibrated parameters (Table 5). For 2023, the model predicts a fertility rate of 1.93, against an actual value of 1.91 (United Nations).

(1) **Gender-neutral productivity growth ($A_t \uparrow$)** raises the opportunity cost of children, reducing both n^s and n^m. Lower n^m reduces men's gains from marriage, pushing $\mathcal{M}$ down. Composition effects further reduce aggregate fertility, female labor supply rises, and Γ^y narrows.

(2) **Gender-biased productivity decline ($B_t \downarrow$)** narrows Γ^w. From women's perspective, a smaller Γ^w lowers the economic value of marriage through $\alpha\Gamma^w$ and reduces $\mathcal{M}$. The same composition mechanism operates, with analogous consequences for n, $l^{♀}$, and Γ^y.

(3) **Dynamic propagation (feedback loop)** operates through (23). A lower $\mathcal{M}_t$ means that more children grow up without both parents, and because boys suffer dispropor-

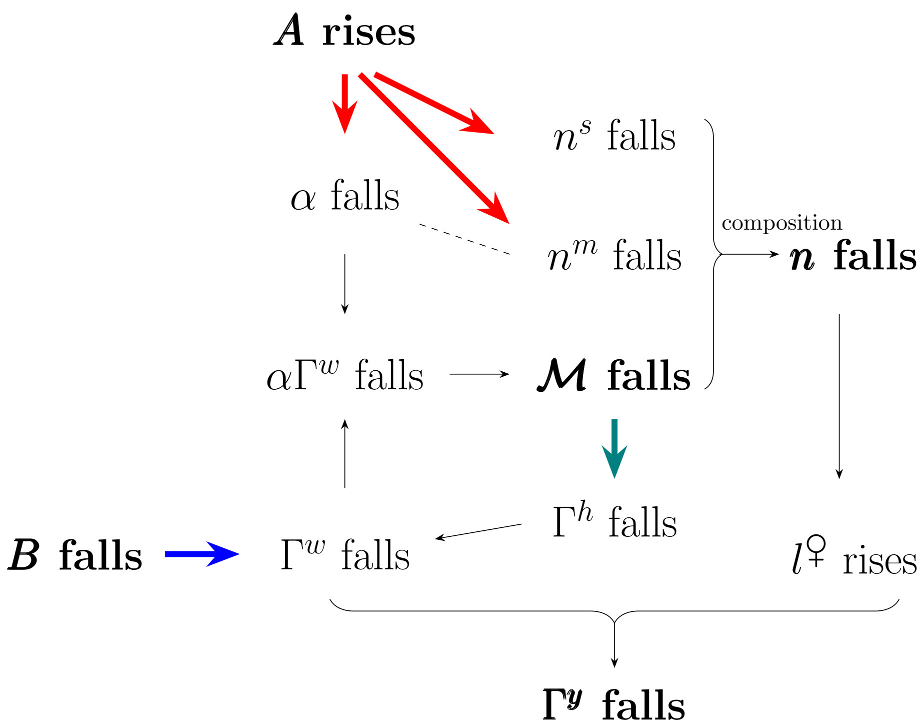


Figure 9. The Dynamic Mechanism

Notes: Red arrows represent the effects of rising gender-neutral productivity A_t. The blue arrow represents the effect of falling gender-biased productivity B_t. The teal arrow represents the dynamic human capital effect through $\mathcal{M}_t \to \Gamma^h_{t+1}$ (Equation (23)).

tionately, Γ^h_{t+1} falls. Less-skilled future men are less marriageable, so next period's $\mathcal{M}$ is lower still, amplifying the decline in n and accelerating Γ^y convergence.

Two implications stand out. First, both gender-neutral and gender-biased technological change can simultaneously depress fertility, erode marriage, and narrow gender income gaps because all three outcomes are tied through the same equilibrium system. This complements earlier emphases on factor-biased technical change in Greenwood et al. (2005b) and Ngai and Petrongolo (2017). Second, when family structure affects boys' and girls' human capital differently, today's fertility and marriage patterns are partly the product of past family structure, and today's shocks echo across generations. Policymakers considering any single outcome must therefore look beyond its immediate spillovers and account for long-run feedback.

6 Conclusion

This paper establishes a three-way trade-off: few societies simultaneously achieve high fertility, widespread dual parenthood, and gender income equality. Across OECD countries, a global panel of 95 nations, and U.S. states and commuting zones, the joint attainment of all three outcomes is a small fraction of what statistical independence would predict. A mutual information analysis shows that the trade-off is genuinely three-dimensional: the three variables exhibit synergy, so no combination of pairwise analyses can recover the joint structure. This statistical finding is the formal justification for studying fertility, marriage, and gender equality jointly rather than pair by pair.

A tractable equilibrium marriage market model explains the pattern. Because marriage bundles an income transfer α with shared fertility n^m, the marriage market clears with a two-dimensional price, and the gender-specific allocation of childcare time ties fertility directly to the gender income gap. Three well-known policy instruments (pro-marriage, gender equality, and overall reductions in childcare costs) each improve two of the three outcomes at the expense of the third. The trade-off, in that sense, binds.

The model's central theoretical contribution is to identify an instrument that can relax the trade-off, namely redistributing childcare from wives to husbands. The redistribution is Pareto-improving within the household when (i) wives' own earnings exceed their spousal transfers, so that shifting the time burden to husbands reduces the wife's effective cost of children, and (ii) husbands value children enough to accept the consumption loss. Both conditions hold comfortably in the calibrated Mexican economy. Paternity leave mandates with reserved father quotas, of the kind implemented in Sweden and Iceland, are the natural policy counterpart.

Quantitatively, the model accounts for Mexico's transformation from 1990 to 2015 using only four estimated parameters and three time-varying wage components. Gender-neutral productivity explains roughly half of the fertility and dual-parenthood decline. Declining gender-biased productivity and human-capital convergence drive most of the gender income and welfare convergence. The 2023 fertility prediction is within 0.02 of the realized value. A dynamic extension, in which boys from single-parent homes accumulate less human capital, shows how a single technological shock can compound into self-reinforcing, cross-generational declines in marriage and fertility.

Fertility, family structure, and gender equality are not three independent policy targets. They are joint outcomes of an equilibrium system in which wages, marriage market clearing, and intra-household time allocation interact. Instruments that operate on the *level* of wages or costs trade one outcome off against the others. Instruments that operate on the *distribution* of childcare can loosen the trade-off, but only in the region of the parameter space identified by conditions (21) and (22). Outside that region, the trade-off binds and policymakers must choose between priorities.

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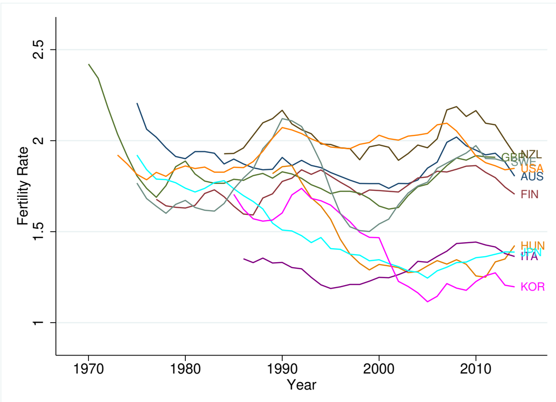
Appendix

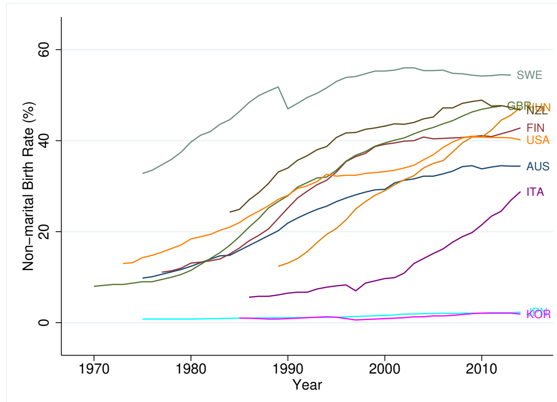
A Additional Empirical Results

A.1 OECD Time-Series Trends

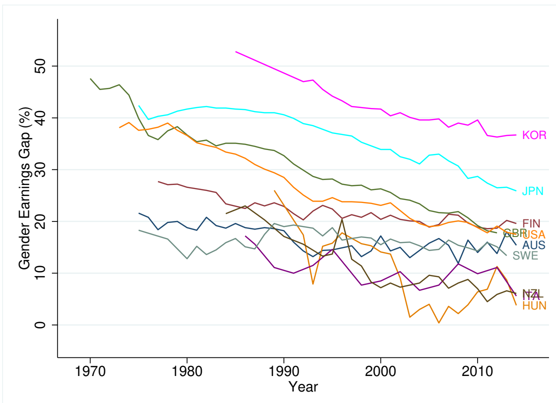
Figure A.1 displays time-series trends in fertility, out-of-wedlock births, and gender earnings gaps for selected OECD countries.



(a) Fertility



(b) Out-of-Wedlock Birth Share



(c) Gender Earnings Gap

Figure A.1. Trends by Country (OECD Sample)

Notes: Time-series trends for OECD countries, 1970 to 2014. Fertility is the total fertility rate (children per woman). The out-of-wedlock birth share is the percentage of births to unmarried parents. The gender earnings gap is defined as (men's median earnings − women's median earnings) divided by men's median earnings.

A.2 OECD Subsample Analysis

Figure A.2 presents Venn diagrams for OECD subsamples stratified by income and human capital levels.

(a) High GDP per Capita

(b) Low GDP per Capita

(c) High Years of Schooling

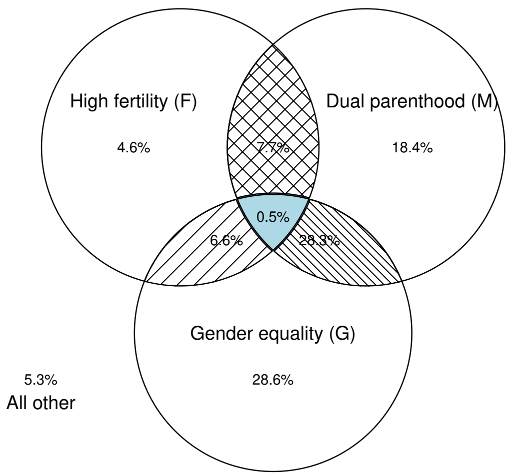
(d) Low Years of Schooling

Figure A.2. Trade-off by Income and Human Capital Levels (OECD Sample)

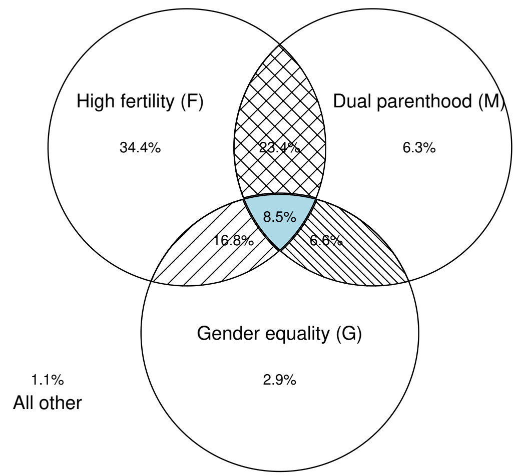
Notes: Venn diagram analysis for subsamples split at median GDP per capita (top row) and median average years of schooling from Barro and Lee (2013) (bottom row). Indicators are redefined at subsample medians in each case. Data cover 37 OECD countries from 1970 to 2014.

A.3 World Subsample Analysis

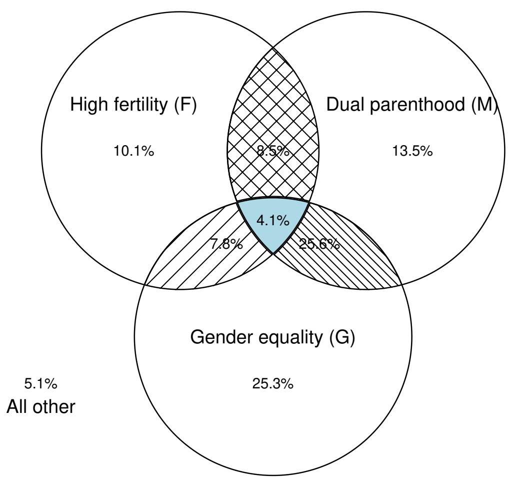
Figure A.3 presents Venn diagrams for World subsamples stratified by income and human capital levels.



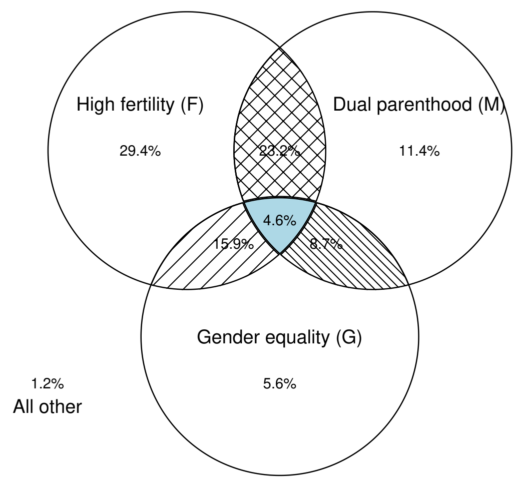
(a) High GDP per Capita



(b) Low GDP per Capita



(c) High Years of Schooling



(d) Low Years of Schooling

Figure A.3. Trade-off by Income and Human Capital Levels (World Sample)

Notes: Venn diagram analysis for subsamples split at median GDP per capita (top row) and median average years of schooling (bottom row). Indicators are redefined at subsample medians in each case. Data cover 95 countries from 1990 to 2015.

A.4 World Sample Transition Paths

Figure A.4 traces transition paths for all countries in the World sample.

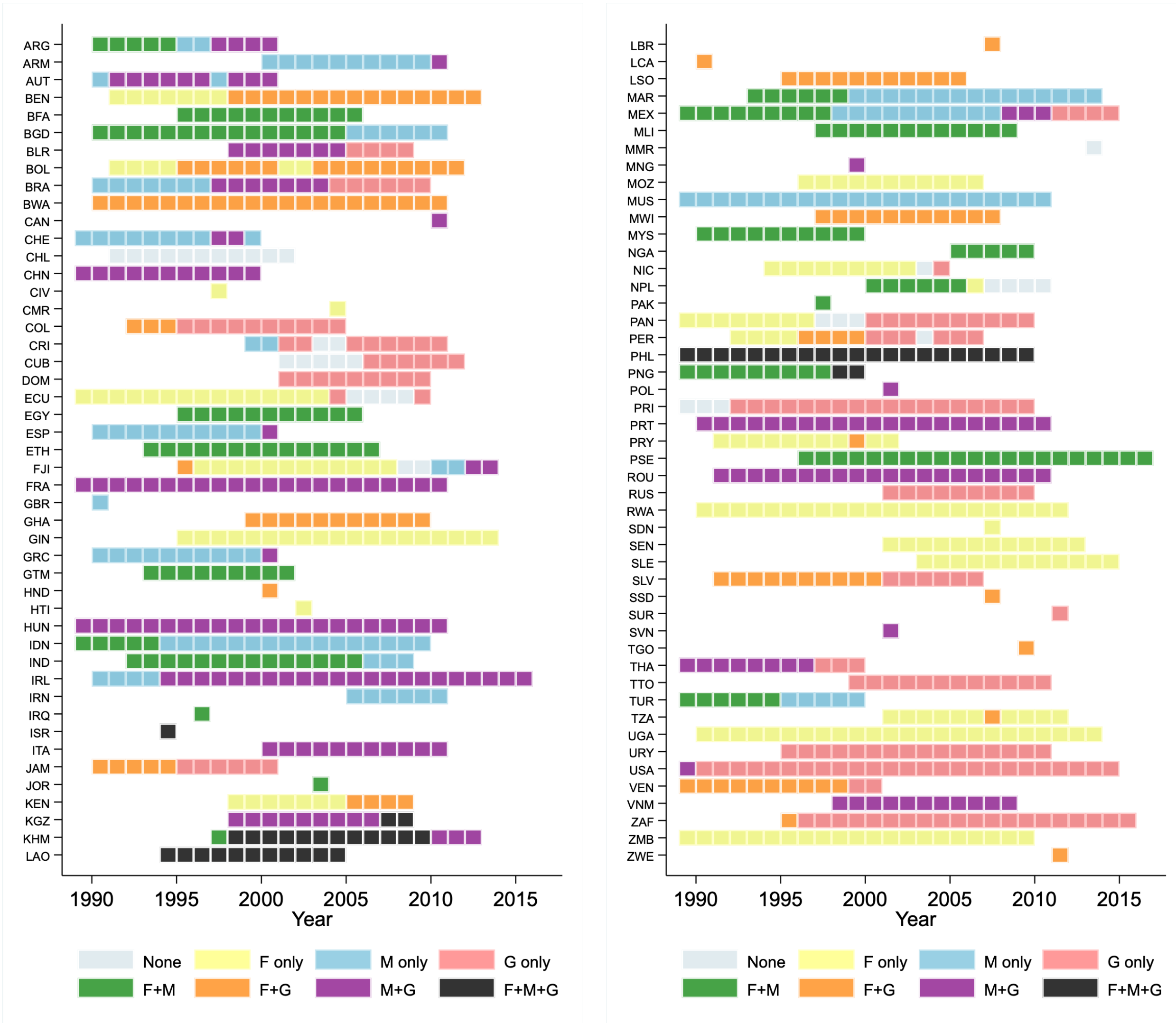


Figure A.4. Transition Paths by Country (World Sample)

Notes: Temporal transitions in the three categories (F = high fertility, M = dual parenthood, G = gender income equality) for all countries in the World sample, 1990 to 2015.

A.5 U.S. Commuting Zone Analysis

Figure A.5 presents the Venn diagram for U.S. commuting zones.

Figure A.6 maps joint attainment by commuting zone.

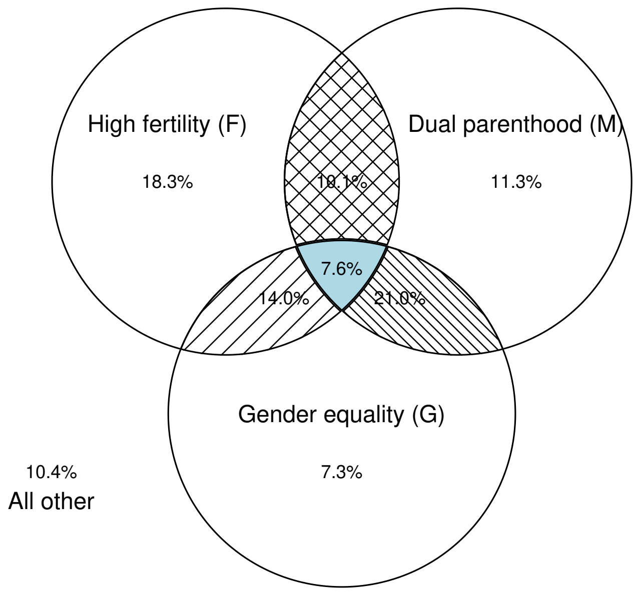


Figure A.5. Trade-off Across U.S. Commuting Zones (2000)

Notes: Venn diagram of high fertility (F: births per 1,000 population > 12.9), dual parenthood (M: single-mother household share < 19.7 percent), and gender earnings equality (G: gender earnings ratio < 1.80), each defined at cross-CZ medians. $N = 741$ commuting zones.

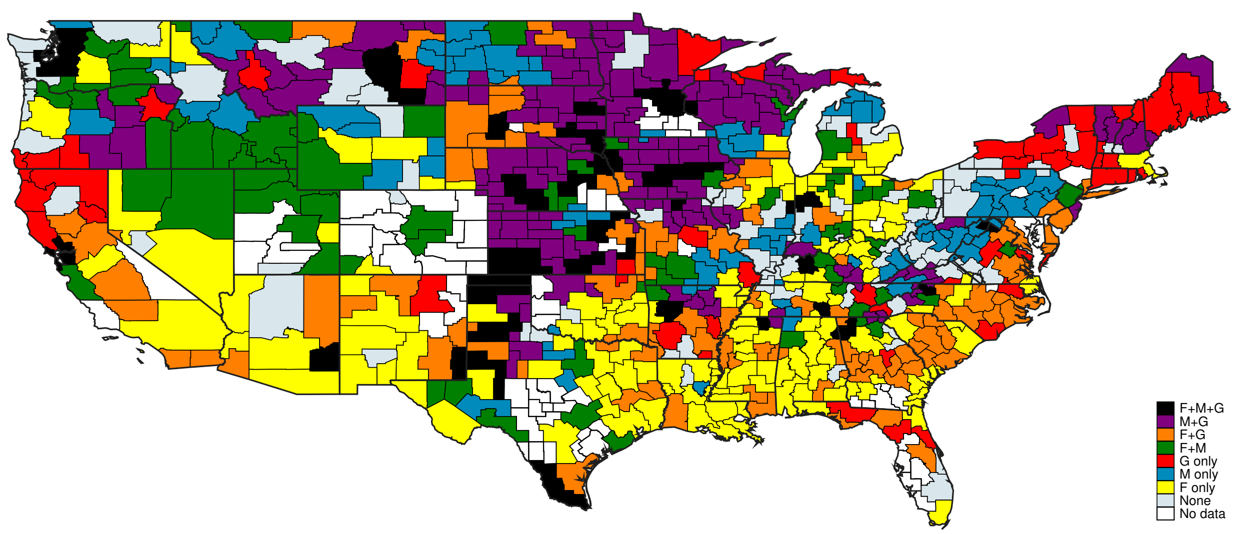


Figure A.6. Joint Attainment by U.S. Commuting Zone (2000)

Notes: The map indicates which outcomes (F = high fertility, M = dual parenthood, G = gender earnings equality) each commuting zone achieves based on median thresholds.

A.6 Summary of Mutual Information Results

Table A.1 provides detailed mutual information results across all samples and subsamples.

Sample	N	Conditional MI			Total Corr.	Interaction
		$I(F;M\|G)$	$I(F;G\|M)$	$I(M;G\|F)$		
OECD						
Full Sample	721	0.078	0.041	0.116	0.165	-0.035^{***}
High Income	360	0.054	0.032	0.129	0.232	$+0.017$
Low Income	361	0.101	0.059	0.108	0.229	-0.039^{**}
High HC	360	0.049	0.041	0.111	0.209	$+0.008$
Low HC	361	0.153	0.087	0.113	0.254	-0.099^{***}
World						
Full Sample	1,130	0.061	0.125	0.035	0.169	-0.026^{***}
High Income	587	0.062	0.092	0.034	0.153	-0.035^{***}
Low Income	543	0.034	0.042	0.027	0.096	-0.007
High HC	613	0.039	0.064	0.024	0.113	-0.014
Low HC	517	0.080	0.102	0.050	0.164	-0.032^{**}
United States						
States	51	0.007	0.089	0.029	0.112	-0.007
CZ	741	0.059	0.015	0.017	0.272	$+0.001$

Table A.1. Detailed Mutual Information Analysis Across Samples

Notes: All values are in bits. F denotes high fertility, M denotes dual parenthood, and G denotes gender equality. Conditional MI reports $I(X;Y|Z)$, the mutual information between X and Y given Z. Total Corr. denotes total correlation. Negative interaction information indicates synergy; positive values indicate redundancy. Statistical significance is based on bootstrap inference with 5,000 replications: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. HC denotes human capital (years of schooling). CZ denotes commuting zones.

B Additional Tables and Figures

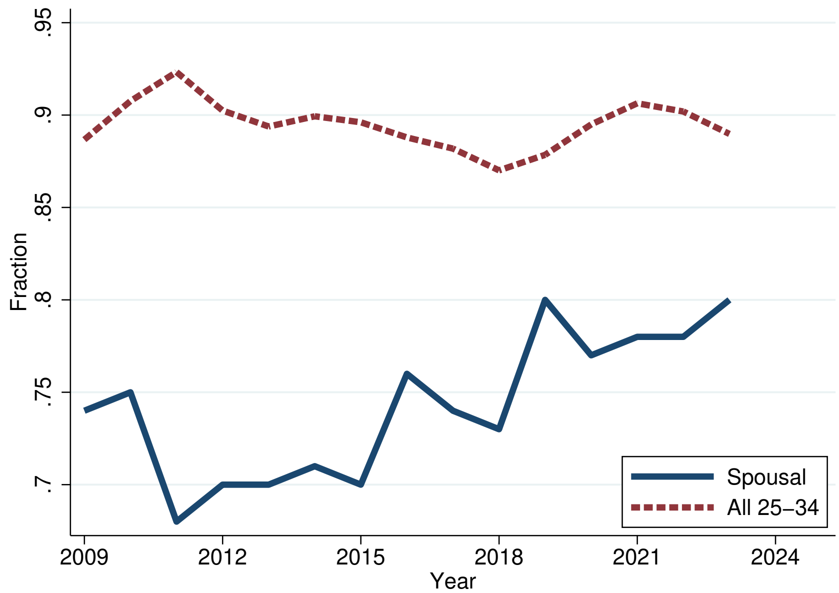
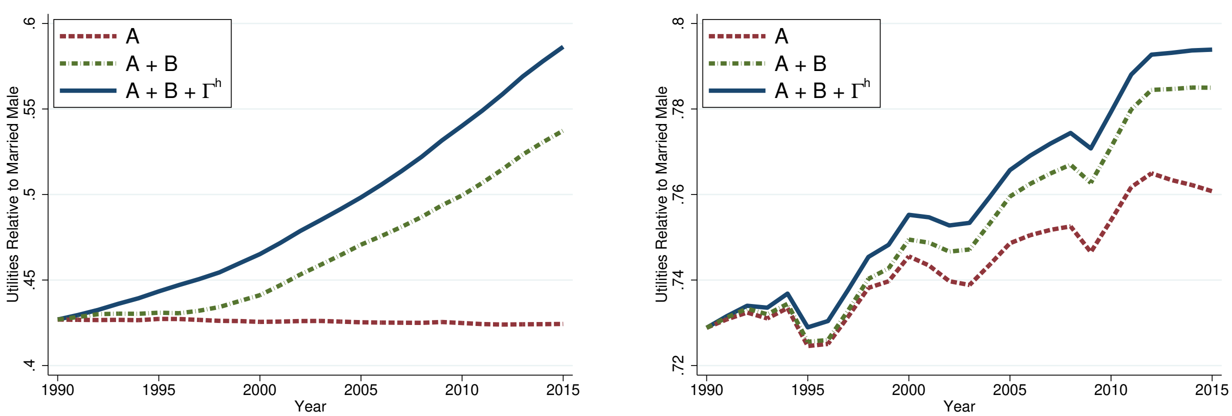


Figure A.7. Gender Earnings Gaps in the U.S.

Notes: The spousal earnings gap is computed using a sample of women from the American Community Survey who are aged 18 to 39, not enrolled in school, childless, and in their first marriage within the past year to a husband who is also not enrolled in school. The overall gender earnings gap for full-time wage and salary workers aged 25 to 34 is drawn from the U.S. Bureau of Labor Statistics' Annual Highlights of Women's Earnings.



(a) Relative Utility: $V^{♀,s}/V^{♂,m}$

(b) Relative Utility: $V^{♂,s}/V^{♂,m}$

Figure A.8. Additional Results on Relative Utility

Notes: Solid lines denote data; dashed lines denote counterfactual paths. "A_t only" varies gender-neutral productivity while fixing B_t and Γ_t^h at their 1990 values. "$A_t + B_t$" allows both to vary while fixing Γ_t^h. "All" refers to the full model.

C Proofs for Baseline Model

C.1 Derivation of Equation (13)

Single men obtain utility $V^{♂,s} = u(w^{♂}, 0) = (1-\beta)^{\rho/(\rho-1)} w^{♂}$. Married men obtain

$$V^{♂,m} = \left[(1-\beta)\big((1-\alpha)w^{♂}\big)^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \cdot \lambda.$$

Taking the ratio and simplifying yields

$$\frac{V^{♂,m}}{V^{♂,s}} = \left[(1-\alpha)^{\frac{\rho-1}{\rho}} + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{♂}}\right)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \cdot \lambda.$$

The marriage market equilibrium condition (7) requires $V^{♂,m}/V^{♂,s} = (\mathcal{M}/(1-\mathcal{M}))^{1/\theta}$. Equating these expressions and solving for α yields equation (13). $\square$

C.2 Proof of Lemma 1

I reduce the two-dimensional fixed-point problem to a single equation in α and exhibit a closed-form solution. To simplify notation, let

$$r \equiv \frac{\rho-1}{\rho} \in (0,1), \qquad C \equiv \left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho}(w^{♀})^{\rho-1} + \chi.$$

For this lemma I replace Assumption 1 with the following condition on primitives, which is strictly weaker in the relevant region of the parameter space and is verified in the calibration.

Condition (A). The primitives satisfy

$$C \cdot w^{♂} > \left(\frac{\beta}{1-\beta}\right)^{\rho/(\rho-1)}. \tag{24}$$

Step 1 (Reduction to a single equation in α). The market-clearing condition $\mathcal{M}^{♂} = \mathcal{M}^{♀}$ in (7) is equivalent, by strict monotonicity of the Fréchet c.d.f., to

$$\frac{V^{♂,m}}{V^{♂,s}} = \frac{V^{♀,m}}{V^{♀,s}}.$$

Using the derivation in Appendix C.1 for the left-hand side and Lemma 2 for the right-hand side, this becomes

$$\left[(1-\alpha)^r + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{\male}}\right)^r\right]^{1/r}\cdot\lambda \;=\; \lambda\,(1+\alpha\Gamma^w).$$

Canceling $\lambda > 0$ and raising both sides to the power $r > 0$,

$$(1-\alpha)^r + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{\male}}\right)^r \;=\; (1+\alpha\Gamma^w)^r. \tag{25}$$

Women's first-order condition (14) pins down n^m as a function of α:

$$n^m(\alpha) \;=\; \frac{1+\alpha\Gamma^w}{C}. \tag{26}$$

Substituting (26) into (25) and factoring out $(1+\alpha\Gamma^w)^r$ on the right yields

$$(1-\alpha)^r \;=\; K\cdot(1+\alpha\Gamma^w)^r, \qquad K \;\equiv\; 1-\frac{\beta}{(1-\beta)\,(Cw^{\male})^r}. \tag{27}$$

Step 2 (Closed-form solution and location). Because $\beta \in (0,1)$, we have $K < 1$ automatically. Condition (A) is equivalent to $(Cw^{\male})^r > \beta/(1-\beta)$, which is equivalent to $K > 0$. Hence $K \in (0,1)$ and $K^{1/r} \in (0,1)$. Both sides of (27) are positive for any $\alpha \in (-1/\Gamma^w, 1)$, so we may take the positive r-th root:

$$1-\alpha \;=\; K^{1/r}\,(1+\alpha\Gamma^w).$$

Solving for α yields the unique closed form

$$\alpha^* \;=\; \frac{1-K^{1/r}}{1+K^{1/r}\Gamma^w}. \tag{28}$$

Since $K^{1/r} \in (0,1)$, both the numerator and denominator are strictly positive, and the numerator is strictly less than the denominator (because $K^{1/r}\Gamma^w > -K^{1/r}$); hence $\alpha^* \in (0,1)$. The corresponding marital fertility is

$$n^{m*} \;=\; \frac{1+\alpha^*\Gamma^w}{C} \;>\; 0.$$

Step 3 (Uniqueness). Every step above is an equivalence on $(0,1)\times\mathbb{R}_{++}$. The map $x \mapsto x^r$ is a strictly increasing bijection on $\mathbb{R}_+$; (26) is a one-to-one linear relation; and (27) admits the unique solution (28) because the function

$$\phi(\alpha) \;\equiv\; (1-\alpha)^r - K\,(1+\alpha\Gamma^w)^r$$

has derivative

$$\phi'(\alpha) \;=\; -r(1-\alpha)^{r-1} - rK\Gamma^w(1+\alpha\Gamma^w)^{r-1} \;<\; 0 \qquad \text{for all } \alpha \in (0,1),$$

so ϕ is strictly decreasing and can cross zero only once. Therefore any equilibrium $(\alpha, n^m) \in (0,1) \times \mathbb{R}_{++}$ must coincide with (α^*, n^{m*}). $\square$

Remark on Condition (A). Condition (A) is extremely mild. Evaluated at the calibrated parameters ($\beta = 0.104$, $\rho = 1.5$, $\chi = 0.075$), the right-hand side of (24) equals $(0.104/0.896)^3 \approx 1.6 \times 10^{-3}$, whereas $Cw^{♂}$ exceeds 0.5 throughout the 1990 to 2015 sample. Condition (A) therefore holds by more than two orders of magnitude. Economically, (24) states that the total cost of raising an additional child (the bracketed term) combined with male labor income is large enough to deliver a strictly positive equilibrium transfer from husbands to wives.

C.3 Proof of Lemma 2

For both married and single women, the first-order condition equating marginal utilities yields the optimal consumption-fertility ratio:

$$\frac{c}{n} = \left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho}.$$

Substituting into the respective budget constraints gives

$$n^m \cdot \left[\left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho} + w^{♀}\chi\right] = (1+\alpha\Gamma^w)w^{♀}, \tag{29}$$

$$n^s \cdot \left[\left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho} + w^{♀}\chi\right] = w^{♀}. \tag{30}$$

Taking the ratio yields

$$\frac{n^m}{n^s} = 1 + \alpha\Gamma^w. \tag{31}$$

Substituting the optimal consumption-fertility ratio into the CES utility function, utility is proportional to fertility:

$$V^{♀,m} = n^m \cdot \Psi \cdot \lambda, \qquad V^{♀,s} = n^s \cdot \Psi,$$

where $\Psi \equiv \left[(1-\beta)\left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho-1}+\beta\right]^{\rho/(\rho-1)}$ is common to both. Therefore

$$\frac{V^{♀,m}}{V^{♀,s}} = \lambda \cdot \frac{n^m}{n^s} = \lambda(1+\alpha\Gamma^w). \quad \square$$

C.4 Proof of Proposition 1

By Lemma 2, women's gains from marriage are $V^{♀,m}/V^{♀,s} = \lambda(1+\alpha\Gamma^w)$. An increase in λ directly raises these gains, which increases the marriage rate $\mathcal{M}$ via equation (7).

Since $n^m > n^s$, a higher $\mathcal{M}$ shifts composition toward married women, raising aggregate fertility $n = \mathcal{M}n^m + (1-\mathcal{M})n^s$.

From (18), higher fertility reduces female labor supply: $l^{♀} = 1-\chi n$ falls. From (17), the gender income gap $\Gamma^y = \Gamma^w/l^{♀}$ therefore widens. $\square$

C.5 Proof of Proposition 2

Step 1: Effect on fertility. From (30), single fertility can be written as

$$n^s = \frac{1}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho}(w^{♀})^{\rho-1}+\chi}.$$

Since $\rho > 1$, the denominator is strictly increasing in $w^{♀}$, so n^s is strictly decreasing in $w^{♀}$.

From (29), marital fertility is

$$n^m = \frac{1+\alpha\Gamma^w}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho}(w^{♀})^{\rho-1}+\chi}.$$

When $w^{♀}$ rises, the numerator falls (since $\Gamma^w = w^{♂}/w^{♀}$ decreases) and the denominator rises, so n^m is strictly decreasing in $w^{♀}$.

Step 2: Effect on marriage. Lower n^m reduces men's gains from marriage and lowers the equilibrium transfer α. By Lemma 2, women's gains $\lambda(1+\alpha\Gamma^w)$ fall through both lower α and lower Γ^w, so $\mathcal{M}$ falls.

Step 3: Aggregate effects. With n^s, n^m, and $\mathcal{M}$ all falling, aggregate fertility n unambiguously declines. Female labor supply $l^{♀} = 1-\chi n$ rises.

Step 4: Gender income gap. The gap is $\Gamma^y = w^{♂}/(w^{♀}l^{♀})$. Since $w^{♀}$ rises and $l^{♀}$ rises, the denominator increases while the numerator is unchanged, so Γ^y falls. $\square$

C.6 Proof of Proposition 3

Step 1: Effect on fertility. From (30),

$$n^s = \frac{1}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho} (w^{♀})^{\rho-1} + \chi}.$$

Taking the derivative with respect to χ (using the chain rule on χ^ρ in the first term and the additive χ in the second) shows $\partial n^s/\partial\chi < 0$. By analogous reasoning, $\partial n^m/\partial\chi < 0$. Both fertilities therefore rise when χ falls.

Step 2: Effect on childcare time. Total childcare time for single women is

$$T^s = \chi n^s = \frac{1}{\left(\frac{(1-\beta)}{\beta}\right)^{\rho} \chi^{\rho-1}(w^{♀})^{\rho-1} + 1}.$$

Since $\rho > 1$, $\partial T^s/\partial\chi < 0$: when χ falls, T^s rises. Similarly, $T^m = \chi n^m$ rises when χ falls.

Step 3: Effect on marriage. Higher n^m raises men's gains from marriage and increases the equilibrium transfer α. By Lemma 2, women's gains rise, so $\mathcal{M}$ increases.

Step 4: Gender income gap. Aggregate childcare time χn rises through three channels: (i) χn^s rises, (ii) χn^m rises, and (iii) composition shifts toward marriage where fertility is higher. Hence $l^{♀} = 1 - \chi n$ falls, and $\Gamma^y = \Gamma^w/l^{♀}$ widens. □

D Extended Model with Gender-Differentiated Childcare Costs

D.1 Setup

Within marriage, let $\chi^{♀}$ and $\chi^{♂}$ denote the time costs per child borne by wives and husbands respectively, with $\chi^{♀} > \chi^{♂}$. Single mothers bear the full cost χ. The budget constraints become

$$c^{♂,m} = (1-\alpha)w^{♂}(1-\chi^{♂}n^m), \tag{32}$$

$$c^{♀,m} = \alpha w^{♂}(1-\chi^{♂}n^m) + w^{♀}(1-\chi^{♀}n^m). \tag{33}$$

Define the wife's *effective income* and *effective childcare cost* as

$$\tilde{w}^{♀} \equiv \alpha w^{♂} + w^{♀}, \qquad \tilde{\chi}^{♀} \equiv \frac{\alpha w^{♂}\chi^{♂} + w^{♀}\chi^{♀}}{\alpha w^{♂} + w^{♀}}.$$

The wife's budget simplifies to $c^{♀,m} = \tilde{w}^{♀}(1-\tilde{\chi}^{♀}n^m)$.

D.2 Equilibrium Fertility

The first-order conditions yield

$$n^s = \frac{1}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho} (w^{♀})^{\rho-1} + \chi}, \qquad n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^{♀}}{\beta}\right)^{\rho} (\tilde{w}^{♀})^{\rho-1} + \tilde{\chi}^{♀}}.$$

D.3 Women's Gains from Marriage

Lemma 3. *In the extended model, the wife's utility from marriage and singlehood can be written as*

$$V^{♀,m} = \lambda \cdot n^m \cdot \Psi^m, \qquad V^{♀,s} = n^s \cdot \Psi^s,$$

where

$$\Psi^m \equiv \left[(1-\beta)\left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho-1} + \beta\right]^{\frac{\rho}{\rho-1}}, \quad \Psi^s \equiv \left[(1-\beta)\left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho-1} + \beta\right]^{\frac{\rho}{\rho-1}}.$$

The wife's gains from marriage are therefore

$$\frac{V^{♀,m}}{V^{♀,s}} = \lambda \cdot \frac{n^m \cdot \Psi^m}{n^s \cdot \Psi^s}.$$

Proof: The optimal consumption-fertility ratio for married women satisfies $c^{♀,m}/n^m = \left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho}$. Substituting into the CES utility function,

$$\begin{aligned} u(c^{♀,m}, n^m) &= \left[(1-\beta)(c^{♀,m})^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \\ &= \left[(1-\beta)(n^m)^{\frac{\rho-1}{\rho}}\left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho-1} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \\ &= n^m \cdot \left[(1-\beta)\left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho-1} + \beta\right]^{\frac{\rho}{\rho-1}} = n^m \cdot \Psi^m. \end{aligned}$$

Thus $V^{♀,m} = \lambda \cdot n^m \cdot \Psi^m$. The derivation for single women is analogous. □

D.4 Proof of Proposition 4

Consider a redistribution that increases $\chi^{♂}$ by δ and decreases $\chi^{♀}$ by δ, holding $\chi^{♂} + \chi^{♀}$ constant.

Part (i). Under the redistribution, the wife's effective childcare cost becomes

$$\tilde{\chi}^{♀}(\delta) = \frac{\alpha w^{♂}(\chi^{♂} + \delta) + w^{♀}(\chi^{♀} - \delta)}{\alpha w^{♂} + w^{♀}}.$$

Differentiating with respect to δ,

$$\frac{\partial \tilde{\chi}^{♀}}{\partial \delta} = \frac{\alpha w^{♂} - w^{♀}}{\alpha w^{♂} + w^{♀}},$$

which is negative if and only if $\alpha w^{♂} < w^{♀}$, that is, condition (21): $\alpha < w^{♀}/w^{♂}$. $\square$

Part (ii). The wife's effective income $\tilde{w}^{♀} = \alpha w^{♂} + w^{♀}$ is unchanged by the redistribution. From the fertility equation

$$n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^{♀}}{\beta}\right)^{\rho} (\tilde{w}^{♀})^{\rho-1} + \tilde{\chi}^{♀}},$$

marital fertility depends on δ only through $\tilde{\chi}^{♀}$. The denominator is strictly increasing in $\tilde{\chi}^{♀}$ (both terms increase), so $\partial n^m/\partial \tilde{\chi}^{♀} < 0$. When condition (21) holds, $\partial \tilde{\chi}^{♀}/\partial \delta < 0$, so by the chain rule

$$\frac{\partial n^m}{\partial \delta} = \frac{\partial n^m}{\partial \tilde{\chi}^{♀}} \cdot \frac{\partial \tilde{\chi}^{♀}}{\partial \delta} = (\text{negative}) \times (\text{negative}) > 0.$$

Single fertility n^s is unchanged since it depends only on χ and $w^{♀}$. $\square$

Part (iii). From Lemma 3, the wife's gains from marriage are $V^{♀,m}/V^{♀,s} = \lambda \cdot (n^m \Psi^m)/(n^s \Psi^s)$. Since n^s and Ψ^s are unaffected by the redistribution, the wife benefits if $\partial(n^m \Psi^m)/\partial \delta > 0$.

Define $A \equiv \left(\frac{1-\beta}{\beta}\right)^{\rho} (\tilde{w}^{♀})^{\rho-1} > 0$, which is constant with respect to $\tilde{\chi}^{♀}$. The fertility equation becomes $n^m = 1/D$ where $D = A(\tilde{\chi}^{♀})^{\rho} + \tilde{\chi}^{♀}$.

From the proof of Lemma 3,

$$\Psi^m = \left[\beta \left(A(\tilde{\chi}^{♀})^{\rho-1} + 1\right)\right]^{\frac{\rho}{\rho-1}}.$$

Noting that $D = \tilde{\chi}^{♀}(A(\tilde{\chi}^{♀})^{\rho-1} + 1)$, we have $A(\tilde{\chi}^{♀})^{\rho-1} + 1 = D/\tilde{\chi}^{♀}$, so

$$\Psi^m = \beta^{\frac{\rho}{\rho-1}} \left(\frac{D}{\tilde{\chi}^{♀}}\right)^{\frac{\rho}{\rho-1}}.$$

The product $n^m\Psi^m$ simplifies to

$$n^m\Psi^m = \frac{1}{D}\cdot\beta^{\frac{\rho}{\rho-1}}\left(\frac{D}{\tilde{\chi}^{♀}}\right)^{\frac{\rho}{\rho-1}} = \beta^{\frac{\rho}{\rho-1}}\cdot D^{\frac{1}{\rho-1}}\cdot(\tilde{\chi}^{♀})^{-\frac{\rho}{\rho-1}}.$$

Define $\xi \equiv A(\tilde{\chi}^{♀})^{\rho-1} > 0$. Then $D = \tilde{\chi}^{♀}(\xi+1)$, so

$$n^m\Psi^m = \beta^{\frac{\rho}{\rho-1}}\cdot(\tilde{\chi}^{♀})^{\frac{1}{\rho-1}}(\xi+1)^{\frac{1}{\rho-1}}\cdot(\tilde{\chi}^{♀})^{-\frac{\rho}{\rho-1}} = \beta^{\frac{\rho}{\rho-1}}\cdot(\tilde{\chi}^{♀})^{-1}\cdot(\xi+1)^{\frac{1}{\rho-1}}.$$

Differentiating with respect to $\tilde{\chi}^{♀}$, using $\partial\xi/\partial\tilde{\chi}^{♀} = (\rho-1)\xi/\tilde{\chi}^{♀}$,

$$\begin{aligned}
\frac{\partial(n^m\Psi^m)}{\partial\tilde{\chi}^{♀}} &= \beta^{\frac{\rho}{\rho-1}}\left[-(\tilde{\chi}^{♀})^{-2}(\xi+1)^{\frac{1}{\rho-1}} + (\tilde{\chi}^{♀})^{-1}\cdot\frac{1}{\rho-1}(\xi+1)^{\frac{1}{\rho-1}-1}\cdot\frac{(\rho-1)\xi}{\tilde{\chi}^{♀}}\right] \\
&= \beta^{\frac{\rho}{\rho-1}}\left[-\frac{(\xi+1)^{\frac{1}{\rho-1}}}{(\tilde{\chi}^{♀})^2} + \frac{\xi(\xi+1)^{\frac{2-\rho}{\rho-1}}}{(\tilde{\chi}^{♀})^2}\right] \\
&= \frac{\beta^{\frac{\rho}{\rho-1}}(\xi+1)^{\frac{2-\rho}{\rho-1}}}{(\tilde{\chi}^{♀})^2}\left[\xi-(\xi+1)\right] \\
&= -\frac{\beta^{\frac{\rho}{\rho-1}}(\xi+1)^{\frac{2-\rho}{\rho-1}}}{(\tilde{\chi}^{♀})^2} < 0.
\end{aligned}$$

The inequality holds because all factors in the final expression are positive for $\rho > 1$. Therefore $\partial(n^m\Psi^m)/\partial\tilde{\chi}^{♀} < 0$ unconditionally.

When condition (21) holds, $\partial\tilde{\chi}^{♀}/\partial\delta < 0$, so

$$\frac{\partial(n^m\Psi^m)}{\partial\delta} = \frac{\partial(n^m\Psi^m)}{\partial\tilde{\chi}^{♀}}\cdot\frac{\partial\tilde{\chi}^{♀}}{\partial\delta} = (\text{negative})\times(\text{negative}) > 0.$$

The wife always benefits from redistribution when condition (21) holds, and no additional restriction on ρ is required. □

Part (iv). The husband's utility from marriage is

$$V^{♂,m} = \lambda\cdot u(c^{♂,m}, n^m) = \lambda\left[(1-\beta)(c^{♂,m})^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}},$$

where $c^{♂,m} = (1-\alpha)w^{♂}(1-\chi^{♂}n^m)$.

Differentiating $V^{♂,m}$ with respect to δ,

$$\frac{dV^{♂,m}}{d\delta} = \lambda\cdot\frac{\rho}{\rho-1}\cdot U^{\frac{1}{\rho-1}}\left[(1-\beta)(c^{♂,m})^{-\frac{1}{\rho}}\frac{dc^{♂,m}}{d\delta} + \beta(n^m)^{-\frac{1}{\rho}}\frac{dn^m}{d\delta}\right],$$

where $U \equiv (1-\beta)(c^{♂,m})^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}} > 0$.

Since $\lambda > 0$, $\rho/(\rho-1) > 0$ (for $\rho > 1$), and $U^{1/(\rho-1)} > 0$, the sign of $dV^{♂,m}/d\delta$ equals the sign of the bracketed term.

Computing $dc^{♂,m}/d\delta$. From the husband's budget constraint,

$$c^{♂,m} = (1-\alpha)w^{♂}(1-\chi^{♂} n^m).$$

Taking the total derivative with respect to δ,

$$\frac{dc^{♂,m}}{d\delta} = (1-\alpha)w^{♂}\left[-n^m\frac{d\chi^{♂}}{d\delta} - \chi^{♂}\frac{dn^m}{d\delta}\right] = (1-\alpha)w^{♂}\left[-n^m - \chi^{♂}\frac{dn^m}{d\delta}\right],$$

where I have used $d\chi^{♂}/d\delta = 1$ (redistribution increases the husband's childcare cost one-for-one with δ).

Since $n^m > 0$, $\chi^{♂} \geq 0$, and $dn^m/d\delta > 0$ (from Part (ii) when condition (21) holds), $dc^{♂,m}/d\delta < 0$: the husband's consumption falls.

Condition for the husband to benefit. The husband benefits ($dV^{♂,m}/d\delta > 0$) if and only if

$$\beta(n^m)^{-\frac{1}{\rho}}\frac{dn^m}{d\delta} > -(1-\beta)(c^{♂,m})^{-\frac{1}{\rho}}\frac{dc^{♂,m}}{d\delta}.$$

Substituting the expression for $dc^{♂,m}/d\delta$,

$$\beta(n^m)^{-\frac{1}{\rho}}\frac{dn^m}{d\delta} > (1-\beta)(c^{♂,m})^{-\frac{1}{\rho}}(1-\alpha)w^{♂}\left[n^m + \chi^{♂}\frac{dn^m}{d\delta}\right].$$

Rearranging,

$$\frac{\beta}{1-\beta}\cdot\left(\frac{c^{♂,m}}{n^m}\right)^{\frac{1}{\rho}}\cdot\frac{dn^m}{d\delta} > (1-\alpha)w^{♂}\left[n^m + \chi^{♂}\frac{dn^m}{d\delta}\right].$$

Dividing both sides by $dn^m/d\delta > 0$,

$$\frac{\beta}{1-\beta}\cdot\left(\frac{c^{♂,m}}{n^m}\right)^{\frac{1}{\rho}} > (1-\alpha)w^{♂}\left[\frac{n^m}{dn^m/d\delta} + \chi^{♂}\right],$$

which is condition (22). □

Comparative statics on condition (22):

- *Effect of β*: The left-hand side is increasing in β, both through $\beta/(1-\beta)$ and indirectly through higher n^m. A higher preference weight on children makes the husband more likely to benefit.

- *Effect of* $\chi^{\male}$: The right-hand side is increasing in $\chi^{\male}$. When the husband already bears a substantial childcare burden, the marginal consumption loss from additional childcare is larger, making him less likely to benefit.

- *Effect of* ρ: The left-hand side involves $(c^{\male,m}/n^m)^{1/\rho}$. Since typically $c^{\male,m} > n^m$ in units, a lower ρ (closer to one) raises the left-hand side, making the husband more likely to benefit. Intuitively, when consumption and children are less substitutable, the marginal utility of additional children is higher.

- *Effect of fertility response*: A larger $dn^m/d\delta$ reduces $n^m/(dn^m/d\delta)$ on the right-hand side, making the condition easier to satisfy. The fertility response is larger when condition (21) holds with greater slack ($\alpha \ll w^{\female}/w^{\male}$).

Parts (v) and (vi). When both spouses benefit from redistribution, each spouse's utility ratio $V^{g,m}/V^{g,s}$ increases. From the marriage market equilibrium condition (7),

$$\mathcal{M}^g = \frac{1}{1 + (V^{g,s}/V^{g,m})^\theta},$$

higher values of $V^{g,m}/V^{g,s}$ for both genders imply a higher equilibrium marriage rate $\mathcal{M}$.

Aggregate fertility is

$$n = \mathcal{M} \cdot n^m + (1 - \mathcal{M}) \cdot n^s.$$

I have established that (i) n^m increases (Part ii), (ii) n^s is unchanged, (iii) $\mathcal{M}$ increases when both spouses benefit (Part v), and (iv) $n^m > n^s$ in equilibrium (married women have higher fertility due to income transfers). All effects work in the same direction, so aggregate fertility n unambiguously increases. □

D.5 Proof of Proposition 5

I treat the two cases in turn. Throughout, I use the notation and intermediate results from the proof of Proposition 4.

Case (A): Condition (21) fails ($\alpha \geq w^{\female}/w^{\male}$). *Step A1 (Effective cost).* From Part (i) of Proposition 4,

$$\frac{\partial \tilde{\chi}^{\female}}{\partial \delta} = \frac{\alpha w^{\male} - w^{\female}}{\alpha w^{\male} + w^{\female}} \geq 0,$$

with strict inequality whenever $\alpha w^{\male} > w^{\female}$.

Step A2 (Marital fertility). Marital fertility $n^m = 1/[A(\tilde{\chi}^{♀})^{\rho} + \tilde{\chi}^{♀}]$ is strictly decreasing in $\tilde{\chi}^{♀}$ because the denominator is strictly increasing in $\tilde{\chi}^{♀}$ (both the $A(\tilde{\chi}^{♀})^{\rho}$ term and the $\tilde{\chi}^{♀}$ term rise). Hence by the chain rule,

$$\frac{\partial n^m}{\partial \delta} \;=\; \frac{\partial n^m}{\partial \tilde{\chi}^{♀}} \cdot \frac{\partial \tilde{\chi}^{♀}}{\partial \delta} \;\leq\; 0.$$

Single fertility n^s is unaffected because it depends only on $w^{♀}$ and χ.

Step A3 (Wife's gains from marriage). The proof of Part (iii) of Proposition 4 establishes the unconditional inequality

$$\frac{\partial (n^m \Psi^m)}{\partial \tilde{\chi}^{♀}} \;<\; 0 \qquad \text{for all } \rho > 1.$$

Combined with Step A1,

$$\frac{\partial (n^m \Psi^m)}{\partial \delta} \;=\; \underbrace{\frac{\partial (n^m \Psi^m)}{\partial \tilde{\chi}^{♀}}}_{<\,0} \cdot \underbrace{\frac{\partial \tilde{\chi}^{♀}}{\partial \delta}}_{\geq\,0} \;\leq\; 0.$$

Since $V^{♀,m}/V^{♀,s} = \lambda \cdot (n^m\Psi^m)/(n^s\Psi^s)$ by Lemma 3, and n^s, Ψ^s are unaffected, the wife's gains from marriage weakly decline.

Step A4 (Marriage rate). The wife's marriage participation $\mathcal{M}^{♀} = 1/[1 + (V^{♀,s}/V^{♀,m})^{\theta}]$ at the initial α weakly declines. Market clearing requires α to adjust so that $\mathcal{M}^{♀} = \mathcal{M}^{♂}$. Because the wife-side participation contracts, the new equilibrium marriage rate weakly declines under standard stability of the matching equilibrium (an upward adjustment of α, which would partially offset the fall in $\mathcal{M}^{♀}$, is bounded by the husband's participation constraint and cannot reverse the direction; see Remark A below).

Step A5 (Aggregate fertility and gender gap). Aggregate fertility is

$$n \;=\; \mathcal{M}\, n^m + (1 - \mathcal{M})\, n^s.$$

In Case (A), n^m weakly falls, $\mathcal{M}$ weakly falls, and $n^m \geq n^s$ in equilibrium, so the composition shift toward singles also reduces n. All three forces work in the same direction: n weakly falls. Female labor supply $l^{♀} = 1 - \chi n$ weakly rises, and the gender income gap $\Gamma^y = \Gamma^w / l^{♀}$ weakly narrows.

Remark A (Husband's response). The husband's consumption is $c^{♂,m} = (1 - \alpha) w^{♂} (1 -$

$\chi^{\male} n^m)$. Differentiating with $d\chi^{\male}/d\delta = 1$,

$$\frac{dc^{\male,m}}{d\delta} \;=\; -(1-\alpha)w^{\male}\left[n^m + \chi^{\male}\,\frac{dn^m}{d\delta}\right].$$

In Case (A) the second term inside the bracket is non-positive (since $dn^m/d\delta \leq 0$), so the bracket can have either sign and the husband's consumption response is ambiguous. The husband's utility response is therefore also ambiguous in general. However, the wife-side contraction in Step A3 is strict whenever Case (A) holds strictly, and the marriage market clears at a (weakly) lower $\mathcal{M}$ regardless.

Case (B): Condition (21) holds but condition (22) fails. Suppose $\alpha < w^{\female}/w^{\male}$ and

$$\frac{\beta}{1-\beta}\left(\frac{c^{\male,m}}{n^m}\right)^{1/\rho} \;<\; (1-\alpha)w^{\male}\left[\frac{n^m}{\partial n^m/\partial\delta} + \chi^{\male}\right].$$

Step B1 (Marital fertility). Part (ii) of Proposition 4 requires only condition (21), so $\partial n^m/\partial\delta > 0$.

Step B2 (Wife's gains). Part (iii) of Proposition 4 requires only condition (21), so $V^{\female,m}/V^{\female,s}$ strictly rises.

Step B3 (Husband's gains). Part (iv) of Proposition 4 establishes that the husband's utility from marriage rises if and only if condition (22) holds. Its failure implies $V^{\male,m}/V^{\male,s}$ strictly falls.

Step B4 (Equilibrium adjustment of α). At the initial α, $\mathcal{M}^{\female}$ strictly rises while $\mathcal{M}^{\male}$ strictly falls, violating $\mathcal{M}^{\female} = \mathcal{M}^{\male}$. To restore market clearing, α must fall: a lower α raises the husband's consumption (and hence $V^{\male,m}$) while reducing the wife's economic surplus $1+\alpha\Gamma^w$ (and hence $V^{\female,m}/V^{\female,s}$ by Lemma 2). The new equilibrium marriage rate $\mathcal{M}^{\dagger}$ satisfies

$$\mathcal{M}^{\male}_{\text{old }\alpha} \;\leq\; \mathcal{M}^{\dagger} \;\leq\; \mathcal{M}^{\female}_{\text{old }\alpha},$$

where the inequalities reflect the partial-equilibrium responses at the initial α. The position of $\mathcal{M}^{\dagger}$ relative to the original equilibrium $\mathcal{M}$ depends on whether the wife-side gain or the husband-side loss dominates the adjustment, and is therefore ambiguous in general.

Step B5 (Aggregate fertility and gender gap). Aggregate fertility $n = \mathcal{M}\,n^m + (1-\mathcal{M})\,n^s$ combines a strictly higher n^m, an ambiguous $\mathcal{M}$, and unchanged n^s. If the marriage rate falls sharply, the composition shift toward singles can dominate and n falls; if the marriage rate falls only modestly, the higher marital fertility dominates and n rises. Female labor supply $l^{\female} = 1 - \chi n$ and the gender income gap $\Gamma^y = \Gamma^w/l^{\female}$ inherit this ambiguity.

Remark B (Where Case (B) is most likely). Inspection of condition (22) shows that it tends to fail in environments with (a) a low preference weight on children (small β), (b) a substantial baseline paternal childcare share (large $\chi^{♂}$), (c) consumption and children that are highly substitutable (large ρ), or (d) a weak fertility response ($\partial n^m/\partial\delta$ small). These features characterize many high-income, low-fertility economies, which the model therefore predicts to be precisely the settings in which paternity-quota reforms generate gender-gap convergence without the accompanying fertility gains. $\square$